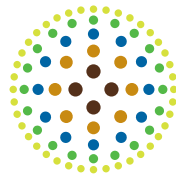




Australian Government

Rural Industries Research and
Development Corporation



RURAL INDUSTRIES

Research & Development Corporation

Potential for the Commercial Culture of the Tropical Sea Urchin *Tripneustes gratilla* in Australia



JULY 2012

RIRDC Publication No. 12/O52



Australian Government

**Rural Industries Research and
Development Corporation**

Potential for the Commercial Culture of the Tropical Sea Urchin *Tripneustes gratilla* in Australia

Benjamin Mos, Kenneth L. Cowden and Symon A. Dworjanyn

July 2012

RIRDC Publication No 12/052
RIRDC Project No PRJ-006543

© 2012 Rural Industries Research and Development Corporation.
All rights reserved.

ISBN 978-1-74254-395-6

Potential for the Commercial Culture of the Tropical Sea Urchin Tripneustes gratilla in Australia
Publication No. 12/052
Project No. PRJ-006543

The information contained in this publication is intended for general use to assist public knowledge and discussion and to help improve the development of sustainable regions. You must not rely on any information contained in this publication without taking specialist advice relevant to your particular circumstances.

While reasonable care has been taken in preparing this publication to ensure that information is true and correct, the Commonwealth of Australia gives no assurance as to the accuracy of any information in this publication.

The Commonwealth of Australia, the Rural Industries Research and Development Corporation (RIRDC), the authors or contributors expressly disclaim, to the maximum extent permitted by law, all responsibility and liability to any person, arising directly or indirectly from any act or omission, or for any consequences of any such act or omission, made in reliance on the contents of this publication, whether or not caused by any negligence on the part of the Commonwealth of Australia, RIRDC, the authors or contributors.

The Commonwealth of Australia does not necessarily endorse the views in this publication.

This publication is copyright. Apart from any use as permitted under the *Copyright Act 1968*, all other rights are reserved. However, wide dissemination is encouraged. Requests and inquiries concerning reproduction and rights should be addressed to the RIRDC Publications Manager on phone 02 6271 4165.

Researcher Contact Details

Symon A Dworjanyn

National Marine Science Centre,
Southern Cross University,
PO Box J321
Coffs Harbour
NSW, 2450

Email: sdworjanyn@nmsc.edu.au

In submitting this report, the researcher has agreed to RIRDC publishing this material in its edited form.

RIRDC Contact Details

Rural Industries Research and Development Corporation

Level 2, 15 National Circuit
BARTON ACT 2600
PO Box 4776
KINGSTON ACT 2604

Phone: 02 6271 4100
Fax: 02 6271 4199
Email: rirdc@rirdc.gov.au.
Web: <http://www.rirdc.gov.au>

Electronically published by RIRDC in July 2012
Print-on-demand by Union Offset Printing, Canberra at www.rirdc.gov.au
or phone 1300 634 313

Foreword

The decline in sea urchin fisheries production has seen increasing interest in the commercial culture of sea urchins. The research presented in this report is important because the diversification of the Australian aquaculture industry into sea urchin aquaculture depends on the development of novel culture methods. The establishment of optimal environmental parameters for maximising growth and production is an important component of the development of these culture methods.

The tropical sea urchin *Tripneustes gratilla* has fast growth rates and high market value, so may be a valuable addition to Australian aquaculture, especially for Australia's tropical North. Any commercial entity interested in modelling the economic viability of the culture of *T. gratilla* will benefit from the research presented here.

This report demonstrates the fast growth rates of *T. gratilla*, particularly at the optimal temperature range of 26-28°C. Additionally, *T. gratilla* maintained high growth, survival and gonad production at high densities and under low seawater exchange rates. However, there was a trade-off between low exchange rates and high densities with production.

Surprisingly, oxygen and nitrogenous wastes were not responsible for limiting growth and gonad production at high densities and low exchange rates. The key environmental parameter that limits growth and gonad production at high densities appears to be a reduction in pH, most likely due to the uptake of carbonate or the production of carbon dioxide by sea urchins.

The information presented in this report provides data to bridge the knowledge gap preventing the establishment of culture methods for *T. gratilla*, and represents an important step forward in the realisation of commercial sea urchin aquaculture in Australia.

This project was funded from RIRDC Core Funds which were provided by the Australian Government.

This report is an addition to RIRDC's diverse range of over 2000 research publications and it forms part of our New Animal Products R&D program, which aims to accelerate the development of viable animal industries.

Most of RIRDC's publications are available for viewing, free downloading or purchasing online at www.rirdc.gov.au. Purchases can also be made by phoning 1300 634 313.

Craig Burns

Managing Director

Rural Industries Research and Development Corporation

Abbreviations

FCE	Food Conversion Efficiency
GI	Gonad Index
LGR	Linear Growth Rate
P_{CO_2}	Partial pressure of carbon dioxide (μatm)
SGR	Specific Growth Rate
TAN	Total Ammonia Nitrogen
TD	Test Diameter
WW	Wet Weight

Contents

Foreword	iii
Abbreviations	iv
Executive Summary.....	viii
Introduction	1
Objectives	2
Methodology.....	3
Study Organisms	3
Experiment 1: Effects of Temperature on Survival and Growth.....	3
Experiment 2: Effects of Culture Density and Seawater Exchange Rate on Juvenile Growth.....	4
Experiment 3: Effects of Culture Density and Seawater Exchange Rate on Adult Growth and Gonad Production.....	6
Experiment 4: Partitioning Sources of Environmental Parameter Change in <i>Tripneustes gratilla</i> culture	7
Statistical Analysis	7
Results.....	8
Experiment 1: The Effects of Temperature on Growth and Survival of <i>Tripneustes gratilla</i>	8
Experiment 2: The Effects of Density and Seawater Exchange Rate on Juvenile Growth and Survival	10
Effect of Density and Seawater Exchange Rate on Juvenile Growth and Survival	10
Effects of Density and Seawater Exchange Rate on Juvenile Consumption and FCE.....	14
Effects of Density and Seawater Exchange Rate on Environmental Parameters	15
Effects of Environmental Parameters on Juvenile Growth	18
Experiment 3: The Effects of Density and Seawater Exchange Rate on Adult Growth and Survival.....	18
Effect of Density and Seawater Exchange Rate on Adult Growth and Survival.....	18
Effects of Density and Seawater Exchange Rate on Adult Gonad Production	22
Effects of Density and Seawater Exchange Rate on Adult Consumption and FCE	22
Effects of Density and Seawater Exchange Rate on Environmental Parameters	24
Effects of Environmental Parameters on Adult Growth and Gonad Production.....	27
Experiment 4: Partitioning Sources of Environmental Parameter Change in <i>Tripneustes gratilla</i> culture.....	28
Discussion	31
Recommendations.....	33
References	34

Tables

Table 1:	Initial densities and seawater exchange rates for a crossed design testing the effects of all combinations of three densities and three exchange rates on the juvenile growth and survival of <i>Tripneustes gratilla</i>	4
Table 2:	Initial densities and seawater exchange rates for a crossed design testing the effects of all combinations of three densities and three exchange rates on the adult growth and survival of <i>Tripneustes gratilla</i>	6
Table 3:	Repeated measures PERMANOVA analyses examining the effects of temperature on (a) cumulative test diameter and (b) cumulative wet weight growth of <i>Tripneustes gratilla</i> over nine weeks.....	10
Table 4:	PERMANOVA analyses examining the effects of density and seawater exchange rate on (a) survival, (b) Specific Growth Rate and (c) Linear Growth Rate of juvenile <i>Tripneustes gratilla</i> after ten weeks	11
Table 5:	PERMANOVA analyses examining the effects of density and seawater exchange rate on (a) consumption and (b) food conversion efficiency (FCE) of <i>Tripneustes gratilla</i> over 20 days in a flow-through seawater system	15
Table 6:	PERMANOVA analyses examining the effects of seawater exchange rate (Flow) and density of <i>Tripneustes gratilla</i> on (a) temperature, (b) nitrate concentration, (c) salinity, (d) pH, (e) carbon dioxide and (f) total alkalinity concentrations in a flow-through seawater system over ten weeks.....	18
Table 7:	PERMANOVA analyses examining the effects of density and seawater exchange rate on (a) Specific Growth Rate, (b) Linear Growth Rate and (c) Gonad Index of adult <i>Tripneustes gratilla</i> after six weeks.	20
Table 8:	PERMANOVA analyses examining the effects of density and seawater exchange rate (Flow) on (a) consumption and (b) food conversion efficiency (FCE) of adult <i>Tripneustes gratilla</i> over seven days in a flow-through seawater system	23
Table 9:	PERMANOVA analyses examining the effects of seawater exchange rate (Flow) and density of adult <i>Tripneustes gratilla</i> on (a) temperature, (b) nitrate concentration, (c) pH, (d) carbon dioxide, (e) total alkalinity and (f) salinity in a flow-through seawater system over six weeks.....	27
Table 10:	PERMANOVA analyses examining the effects of seawater exchange rate and presence of <i>Sargassum</i> spp., <i>Tripneustes gratilla</i> and <i>T. gratilla</i> faeces on (a) nitrate concentration, (b) pH, (c) total alkalinity and (d) carbon dioxide concentration (P_{CO_2}) in a flow-through seawater system	30

Figures

Figure 1:	Juvenile <i>Tripneustes gratilla</i> cultured at the National Marine Science Centre, Southern Cross University.....	1
Figure 2:	Experimental apparatus used to test the effect of four temperature regimes on growth and survival of <i>Tripneustes gratilla</i>	3
Figure 3:	Experimental apparatus used to test the effects of all combinations of three seawater exchange rates and three culture densities on growth, survival and gonad production of <i>Tripneustes gratilla</i>	5
Figure 4:	Effect of temperature on the growth of <i>Tripneustes gratilla</i> over nine weeks. (a) Cumulative test diameter. (b) Cumulative wet weight. (c) Mean test size and representative spine and test colour variance.....	9
Figure 5:	Effect of density and seawater exchange rate (Flow) on survival of <i>Tripneustes gratilla</i> over ten weeks in a flow-through seawater system.....	10
Figure 6:	Effects of density and seawater exchange rate (Flow) on the growth of <i>Tripneustes gratilla</i> over ten weeks in a flow-through seawater system. (a) Cumulative wet weight. (b) SGR (Specific Growth Rate). (c) Cumulative test diameter. (d) LGR (Linear Growth Rate).....	12
Figure 7:	Representative size and densities of juvenile <i>Tripneustes gratilla</i> after ten weeks in a flow-through culture system under different density and seawater exchange rate regimes.	13
Figure 8:	Effects of density and seawater exchange rate (Flow) on (a) consumption and (b) food conversion efficiency (FCE) of juvenile <i>Tripneustes gratilla</i> over 20 days in a flow-through seawater system.....	14
Figure 9:	The effects of seawater exchange rate (Flow) and density of <i>Tripneustes gratilla</i> on environmental parameters in a flow-through seawater system over ten weeks. (a) Temperature. (b) Nitrate concentration. (c) Salinity.....	16
Figure 10:	Effects of density and seawater exchange rate on growth of adult <i>Tripneustes gratilla</i> after six weeks in a flow-through seawater system. a) Cumulative wet weight. (b) SGR (Specific Growth Rate). (c) Cumulative test diameter. (d) LGR (Linear Growth Rate)...	19
Figure 11:	Representative size and densities of adult <i>Tripneustes gratilla</i> after six weeks in a flow-through culture system under different density and seawater exchange rate regimes.	21
Figure 12:	Effect of density and seawater exchange rate (Flow) on gonad indices of <i>Tripneustes gratilla</i> after six weeks in a flow-through seawater system.....	22
Figure 13:	Effects of density and seawater exchange rate (Flow) on (a) consumption and (b) food conversion efficiency (FCE) of adult <i>Tripneustes gratilla</i> over seven days in a flow-through seawater system. Bars denote statistical differences in consumption between treatments (Table 8a; followed by post-hoc pair-wise test, $P < 0.05$).....	23
Figure 14:	The effects of seawater exchange rate (Flow) and density of adult <i>Tripneustes gratilla</i> on environmental parameters in a flow-through seawater system over six weeks. (a) Temperature. (b) Nitrate concentration. (c) pH.....	25
Figure 15:	The effects of seawater exchange rate and presence of <i>Sargassum</i> spp., <i>Tripneustes gratilla</i> and <i>T. gratilla</i> faeces on (a) nitrate concentration, (b) pH, (c) total alkalinity and (d) carbon dioxide concentration (P_{CO_2}) in a flow-through seawater system. Combined = Faeces + Sargassum + Urchins, Control = Seawater only. Capitalised letters denote statistical differences between treatments.....	29

Executive Summary

What the report is about

The decline in sea urchin fisheries production has seen increasing interest in the commercial culture of sea urchins, including the fast-growing tropical *Tripneustes gratilla*. However, the establishment of commercial sea urchin culture ventures is hampered by a lack of data on optimal culture conditions for sea urchins. The information presented in this report provides data to bridge this knowledge gap and represents an important step forward in the realisation of commercial sea urchin aquaculture.

Who is the report targeted at?

This report is targeted at commercial entities interested in diversifying into sea urchin aquaculture. Information on optimal growth, survival and gonad production is required in order to model the economics of sea urchin aquaculture ventures in Australia. This report also contributes to the growing global literature on sea urchin aquaculture, and may therefore be of interest to aquaculture researchers.

Where are the relevant industries located in Australia?

At present, there are no commercial sea urchin aquaculture ventures in Australia. However, there is substantial interest within the aquaculture industry in diversification into sea urchin aquaculture.

Background

The continuing decline in the productivity of the global sea urchin fishery has seen increasing demand for sea urchin products. This demand is driving the development of culture methods for several commercially important sea urchin species. Although the lifecycles of some sea urchin species have been closed, there remain substantial gaps in the knowledge of how environmental parameters affect survival and growth of sea urchins in culture. Establishing optimal culture conditions to maximise growth, survival and gonad production is an imperative requirement for efficient and profitable commercial sea urchin aquaculture.

Tripneustes gratilla is a commercially valuable tropical sea urchin with high growth rates. It is being targeted as an aquaculture species in several countries, including Australia. However, the economic assessment and establishment of aquaculture ventures for the commercial production of *T. gratilla* is hampered by a lack of data on optimal culture conditions for juvenile and adult growth.

Aims/objectives

The objective of this study was to investigate the effects of culture density, seawater exchange rate, temperature, nitrogenous wastes and dissolved gases (oxygen, carbon dioxide) on growth, survival and gonad production on the juvenile and adult growth phases respectively for the tropical sea urchin *T. gratilla*.

Methods used

This study was divided into four experiments; each examined utilising pilot-scale sea water systems. The first examined the effect of temperature on survival and growth of *T. gratilla*. The optimal temperature for growth and survival demonstrated by this experiment was utilised in following experiments.

The second experiment examined the effects of culture density and seawater exchange rate on growth and survival of juvenile sea urchins. Oxygen, carbon dioxide, nitrogenous wastes, total alkalinity and salinity levels were monitored and were statistically compared to growth rates of the urchins to establish which of these environmental parameters may have been driving the patterns seen in growth.

The third experiment utilised the design of the second experiment, except examined the growth, survival and gonad production of adult sea urchins.

Finally, as environmental parameters (water quality) were different between treatments in the preceding two experiments, an experiment was done to investigate how the sea urchins, their food and faeces influenced water quality under different seawater exchange rates, and how this varied between night and day.

Results/key findings

This report demonstrates the fast growth rates of *T. gratilla*, particularly at the optimal temperature range of 26-28°C. Additionally, *T. gratilla* maintained high growth, survival and gonad production at high densities and under low seawater exchange rates. However, there was a trade-off between low exchange rates and high densities with production. These results provide data whereby an economic model for specific aquaculture systems could be constructed.

Surprisingly, oxygen and nitrogenous wastes did not limit growth and gonad production at high densities and low exchange rates. The key environmental parameter that limits growth and gonad production at high densities appears to be a reduction in pH, most likely due to the uptake of carbonate or the production of carbon dioxide by the sea urchins.

Implications for relevant stakeholders:

The rapid growth rates and tolerance to high culture densities of *T. gratilla* singles this species out as an attractive prospect for culture. The relatively high optimal growth temperature suggests that it will be particularly attractive for culture in tropical areas of Australia. This report provides data that is essential for economic assessment of the commercial production of *T. gratilla*. However, any such analysis would need to incorporate site specific information and relative pricing of seawater exchange rates, tank space and market prices for gonads.

Recommendations

This report recommends *T. gratilla* as an ideal candidate for aquaculture. Its exceptionally high growth rates and ability to maintain high growth, survival and gonad production under extremely high densities makes it an attractive proposition.

It is recommended that the optimum culture temperature for this species is between 26-28°C.

It is recommended that any prospective producer of *T. gratilla* should use the data in this report to construct models to investigate the economic viability of such an operation. Any such analysis would need to incorporate site specific information and relative pricing of seawater exchange rates, tank space and market prices for gonads. Therefore, this report does not make any specific recommendations regarding these parameters.

This report further recommends that several important hurdles to the commercial production of *T. gratilla* remain. These include:

- A bottle-neck in the survival of newly settled juveniles (see Mos et al., 2011).
- Establishment of protocols to allow uniform gonad conditioning of adults prior to harvest.
- Creation and production of cost-effective diets that optimise growth.

These problems will need to be addressed before the establishment of a sea urchin aquaculture industry in Australia can occur.

Introduction

The continuing decline in the productivity of the global sea urchin fishery (Andrew et al., 2002; UNFAO, 2012) has seen increasing demand for sea urchin products. This demand is driving the development of culture methods for several commercially important sea urchin species. Although the lifecycles of some sea urchin species have been closed (e.g. Daggett et al., 2006; Devin, 2002; Grosjean et al., 1998; Shimabukuro, 1991), there remain substantial gaps in the knowledge of how environmental parameters affect survival and growth of sea urchins in culture. Establishing optimal culture conditions to maximise growth, survival and gonad production is an imperative requirement for efficient and profitable commercial sea urchin aquaculture.

Somatic growth of juveniles is an important phase in sea urchin culture because it is often long (Basuyaux & Blin, 1998; Pearce et al., 2005), requiring substantial resource investment. Optimising growth and survival through the manipulation of environmental parameters should increase efficiency and minimise culture time during this phase, but has received little attention for most sea urchin species (Azad et al., 2010). Temperature is one of the most important factors affecting the growth of sea urchins, but optimal temperatures for juvenile growth have only been determined for *Strongylocentrotus* spp. (Devin et al., 2004; Miller & Emlet, 1999; Pearce et al., 2005). Similarly, the effects of density remain unknown for most species (Kelly, 2002). The effects of diet on juvenile growth has received comparatively greater attention for several species (Basuyaux & Blin, 1998; Daggett et al., 2005; Dworjanyn et al., 2007; Kelly, 2002; McBride et al., 1998).

Establishing the effects of environmental parameters on growth, survival and gonad production of adult sea urchins has received comparatively greater attention than that for juveniles. Growth, survival and gonad production of adult sea urchins are affected by density, temperature, diet, photoperiod, ammonia, nitrite, dissolved oxygen and carbon dioxide concentration (Azad et al., 2011; Bottger et al., 2006; Cook et al., 1998; Hammer et al., 2006; Juinio-Menez et al., 2008; Pearce et al., 2002a,b; Siikavuopio et al., 2004a,b; 2007a,b,c; 2008). It is likely that these parameters will also affect juvenile growth as sea urchin physiology is not substantially different between juveniles and adults. However, sea urchins allocate resources differently within these two growth phases so assessment of these and other, parameters will need to be determined separately. Additionally, factors affecting growth and survival of sea urchins should be investigated in concert, as single factor experiments may miss important interactions that may be limiting growth and survival (Siikavuopio et al., 2004a).



Figure 1: Juvenile *Tripneustes gratilla* cultured at the National Marine Science Centre, Southern Cross University. Source: H. Sheppard Brennand.

Tripneustes gratilla (Fig. 1) is a commercially valuable tropical sea urchin with high growth rates (Dworjanyn et al., 2007; Lawrence & Agatsuma, 2001). It is being targeted as an aquaculture species in several countries, including Australia (Dworjanyn et al., 2007; Juinio-Menez et al., 2008; Mos et al., 2011; Shimabukuro, 1991), is produced in small quantities for restocking in Japan (Andrew et al., 2002; Shimabukuro, 1991) and as food in the Philippines (Juinio-Menez et al., 2008).

Australia is well placed to take advantage of the high productivity of *T. gratilla* given its 'clean-green' image, large coastal areas available for aquaculture, the availability of innovative aquaculture technology and established commercial ties with the Japanese seafood market. However, the economic assessment and establishment of aquaculture ventures for the commercial production of *T. gratilla* is hampered by a lack of data on optimal culture conditions for juvenile and adult growth.

Objectives

The objective of this study was to investigate the effects of culture density, seawater exchange rates, temperature, nitrogenous wastes and dissolved gases (oxygen, carbon dioxide) on growth, survival and gonad production on the juvenile and adult growth phases respectively for the tropical sea urchin *T. gratilla*.

To achieve this objective, four experiments were conducted. The first examined the effect of temperature on survival and growth of *T. gratilla*. The optimal temperature for growth and survival demonstrated by this experiment was utilised in following experiments.

The second experiment examined the effects of culture density and seawater exchange rate on growth and survival of juvenile sea urchins. Oxygen, carbon dioxide, nitrogenous wastes, total alkalinity and salinity levels were monitored, and were statistically compared to growth rates of the urchins to establish which of these environmental parameters may have been driving the patterns seen in growth.

The third experiment utilised the design of the second experiment, except examined the growth, survival and gonad production of adult sea urchins.

Finally, as environmental parameters (water quality) were different between treatments in the preceding two experiments, an experiment was done to investigate how the sea urchins, their food and faeces influenced water quality under different seawater exchange rates and how this varied between night and day. A fully crossed design was used to partition the effects of light, two seawater exchange rates and three possible sources/sinks (sea urchins, faeces and seaweed) on nitrate concentration, pH, total alkalinity and carbon dioxide concentration.

Methodology

Study Organisms

T. gratilla were obtained from cultures conducted at the National Marine Science Centre, Southern Cross University, Coffs Harbour, Australia (30°12.5'S, 153°16.1'E). Larval culture methods were as described by Mos et al., (2011). Larvae were induced to settle using naturally derived biofilms and *Sargassum* spp. conditioned seawater (Mos et al., 2011). Settled juveniles fed on the biofilm post-settlement until a test diameter (hereafter TD) of 5-10mm when macroalgae (*Sargassum* spp.) was introduced.

Experiment 1: Effects of Temperature on Survival and Growth

To test the effect of temperature on the growth and survival of *T. gratilla*, urchins were exposed to four temperature treatments (19-21°C, 23-25°C, 26-28°C and 29-31°C) over a period of nine weeks. Each treatment had nine replicates, consisting of a 500mL round plastic container with a meshed window that maintained the water level at 400mL. Each replicate contained one randomly assigned sea urchin. The urchins were starved for two weeks prior to use in the experiment to standardise their nutritional condition, and during the experiment were fed *Sargassum* spp. *ad libitum*. The treatments were maintained on a photoperiod of 12:12 (light: dark) supplied by ambient fluorescent lighting.

Each replicate was supplied with preheated seawater at an initial rate of 2.5L.hr⁻¹, increasing to 10L.hr⁻¹ by the end of the experiment. Seawater was heated to the respective treatment temperatures using thermostat controlled aquarium heaters in 60L header tanks (Fig. 2). Temperatures in each replicate were measured daily. Replicate containers were initially cleaned of faeces weekly, but increased to four times per week by the end of the experiment.

To assess somatic growth, TD and wet weight (hereafter WW) measurements were taken weekly. TD was assessed from digital photographs using publically available ImageJ image analysis software (NIH, USA). Two perpendicular measurements of diameter at the longest axis were taken for each sea urchin and the average used for statistical analysis. WW was assessed to the nearest 0.01g after removal of excess sea water by positioning urchins on dry paper towel on anterior and posterior ends for 15 seconds each.

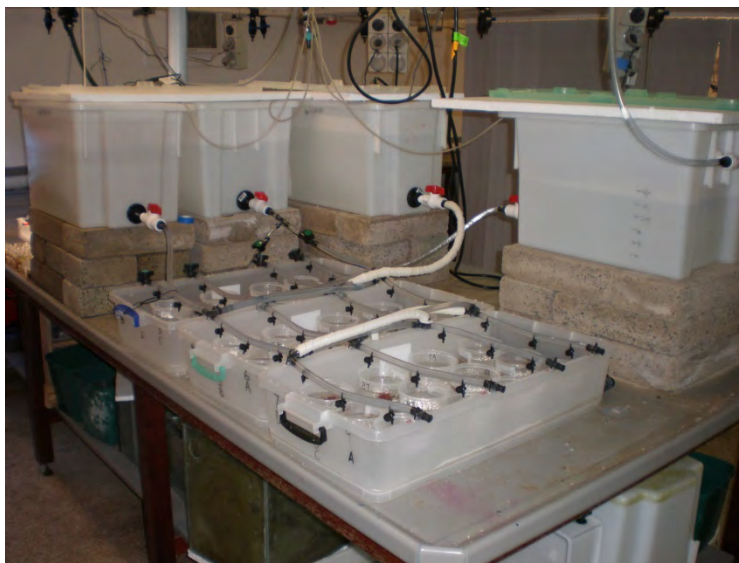


Figure 2: Experimental apparatus used to test the effect of four temperature regimes on growth and survival of *Tripneustes gratilla*. Source: B. Mos.

Experiment 2: Effects of Culture Density and Seawater Exchange Rate on Juvenile Growth

A fully crossed design was used to test the effects of all combinations of three densities (Low, Medium, High) and three seawater exchange rates (Low, Medium, High) on the juvenile growth of *T. gratilla* over ten weeks (Table 1). Each treatment had five replicates consisting of a 5L round plastic bucket (180mm Øbase, 200mm H) with over-flow holes that maintained the water level at 4L. Replicate buckets were randomly assigned positions in a water bath held at a depth of 100mm to maintain stable water temperatures. Each replicate was individually aerated using a single airstone. Dissolved oxygen was maintained above 6.00mg.L⁻¹ (>90.0%) in all replicates, checked weekly using a Hach® HQ40d multi-controller and a Hach® LDO101 probe. Seawater was filtered to 20µm and supplied from an 800L insulated header tank, heated (~27°C) using a flow-through 3000W heater (Elecro Engineering TO3 Ti). Seawater flow to each replicate (flow-through) was regulated using irrigation drip taps (Fig. 3). The treatments were maintained on a photoperiod of 12:12 (light: dark) supplied by ambient fluorescent lighting.

T. gratilla were starved for two weeks prior to use in the experiment to standardise their nutritional condition, and during the experiment were fed *Sargassum* spp. to excess. Replicates were examined daily for mortalities. TD and WW were assessed weekly as previously described, during which replicate buckets were cleaned of faeces. Results were expressed as SGR (Specific Growth Rate) and LGR (Linear Growth Rate) calculated by:

$$\text{SGR}\% = (\ln W_f - \ln W_i \times 100) / t_f - t_i$$

$$\text{LGR} = (L_f - L_i) / t_f - t_i$$

where W_i and W_f were the initial and final mean WW(g) respectively of the urchins, L_i and L_f were the initial and final mean TD(mm) and t_i and t_f were the initial and final time points (weeks).

Table 1: Initial densities and seawater exchange rates for a crossed design testing the effects of all combinations of three densities and three exchange rates on the juvenile growth and survival of *Tripneustes gratilla*

Density	kg.m⁻³	Individuals. m⁻² (Bottom)	Individuals. m⁻² (Total)	Exchange Rate	Exchanges per hour
Low	5	196	43	Low	0.3
Medium	10	392	86	Medium	1.0
High	15	588	129	High	3.0

Daily measurements of temperature, pH and salinity (Hach® HQ40d multi-controller, Hach® PHC101 pH probe, Hach® CDC101 conductivity probe) were recorded for each replicate and the ambient seawater supplied to the experiment. Total Ammonia Nitrogen (hereafter TAN), nitrite and nitrate concentrations were photometrically assessed from water samples collected daily (Palintest® photometer 7100, Ammonia AP152, Nitricol AP109 and Nitratetest AP163). Samples were pooled from each treatment and from the ambient seawater supplied to the experiment. Total alkalinity was assessed via potentiometric titration (Metrohm 888 Titrando) from water samples pooled from each treatment and the ambient seawater supplied to the experiment, collected weekly. P_{CO_2} concentrations were calculated from temperature, pH, salinity, and total alkalinity data using CO2SYS (Pierrot et al., 2006).

Consumption rates for each treatment were documented from week six to week eight of the experiment. The initial and final WW of *Sargassum* spp. fed to each replicate, and the elapsed time (minutes), was recorded for approximately 24 hours for a period of 20 days. Mean autogenic change

in WW of *Sargassum* spp. over 24 hours recorded in a pilot trial (n=30) was less than 1.0%, therefore autogenic change in WW of *Sargassum* spp. during the feeding experiment was assumed to be negligible. Consumption rates were calculated by:

$$\text{Consumption} = [(S_i - S_f) / d_i] / (t_f - t_i)$$

where S_i and S_f were the initial and final WW respectively of the *Sargassum* spp., d_i was the initial mean WW of the urchins and t_f and t_i were the final and initial time points respectively.

FCE (Food Conversion Efficiency) was calculated by the formula:

$$\text{FCE\%} = [(d_f - d_i) / S] * 100$$

where d_i and d_f were the initial and final mean WW respectively of the urchins and S was the total WW of *Sargassum* spp. consumed.



Figure 3: Experimental apparatus used to test the effects of all combinations of three seawater exchange rates and three culture densities on growth, survival and gonad production of *Tripneustes gratilla*. Source: B. Mos.

Experiment 3: Effects of Culture Density and Seawater Exchange Rate on Adult Growth and Gonad Production

A fully crossed design was used to test the effects of all combinations of three densities (Low, Medium, High) and three seawater exchange rates (Low, Medium, High) on the adult growth, survival and gonad production of *T. gratilla* over six weeks (Table 2). Each treatment had five replicates consisting of a 5L round plastic buckets (previously described), randomly assigned positions in a water bath held at a depth of 100mm to maintain stable water temperatures. Each replicate was individually aerated using a single airstone. Dissolved oxygen was maintained above 6.00mg.L⁻¹ (>90.0%) in all replicates, checked weekly. Seawater was filtered to 20µm and supplied from an 800L insulated header tank, heated (~27°C) using a flow-through 3000W (Watt) heater (Elecro Engineering TO3 Ti). Seawater flow to each replicate (flow-through) was regulated using irrigation drip taps. The treatments were maintained on a photoperiod of 12:12 (light: dark) supplied by ambient fluorescent lighting.

Table 2: Initial densities and seawater exchange rates for a crossed design testing the effects of all combinations of three densities and three exchange rates on the adult growth and survival of *Tripneustes gratilla*

Density	kg.m⁻³	Individuals. m⁻² (Bottom)	Individuals. m⁻² (Total)	Exchange Rate	Exchanges per hour
Low	27	39	9	Low	0.3
Medium	81	118	26	Medium	1.0
High	135	196	43	High	3.0

T. gratilla were starved for two weeks prior to use in the experiment to standardise their nutritional condition, and during the experiment were fed *Sargassum* spp. to excess. Replicates were examined daily for mortalities. TD and WW were assessed weekly as previously described, during which replicate buckets were cleaned of faeces. Results were expressed as SGR and LGR.

Daily measurements of temperature, pH and salinity were recorded for each replicate and the ambient seawater supplied to the experiment. TAN, nitrite and nitrate concentrations were photometrically assessed from water samples collected daily. Samples were pooled from each treatment and from the ambient seawater supplied to the experiment. Total alkalinity was assessed via potentiometric titration from water samples pooled from each treatment and the ambient seawater supplied to the experiment, collected weekly. *P*_{CO2} concentrations were calculated from temperature, pH, salinity, and total alkalinity data using CO2SYS (Pierrot et al., 2006).

To establish the mean gonad condition at the start of the experiment, gonads were removed from 25 urchins and were weighed to the nearest 0.01g after removal of excess sea water by positioning gonads on dry paper towel for 15 seconds each side. A gonad index (GI) was calculated for each urchin using the formula:

$$GI = [G_{WW} / WW] * 100$$

where *G_{WW}* was the wet weight of the gonads and *WW* was the total *WW* of the sea urchin.

This procedure was repeated for all urchins in the experiment after six weeks to establish mean gonad indices for each treatment.

Consumption rates for each treatment were documented during week four of the experiment. The initial and final *WW* of *Sargassum* spp. fed to each replicate, and the elapsed time (minutes), was recorded for approximately 24 hours for a period of seven days. FCE was calculated from this data as previously described.

Experiment 4: Partitioning Sources of Environmental Parameter Change in *Tripneustes gratilla* culture

As pH, P_{CO_2} , total alkalinity and nitrate concentrations were significantly different between seawater exchange rates (see Results), an overnight trial was conducted to partition factors that could influence these environmental conditions. A fully crossed design was used to test the effect of all combinations of light (Light vs. Dark), seawater exchange rate (High vs. Low, Table 1) and three treatments, *T. gratilla* only (15 individuals per replicate, $\sim 200\text{kg}\cdot\text{m}^{-3}$), *Sargassum* spp. only, *T. gratilla* faeces only, and a control combination of *T. gratilla*, *Sargassum* spp. and faeces. Each treatment had five replicates, consisting of the 5L buckets previously described.

The experiment was conducted in the seawater system previously described (Fig. 2). The treatments were maintained on a photoperiod of 12:12 (light: dark) supplied by ambient fluorescent lighting. Seawater samples were collected from each replicate after 24hrs during the light phase of the photoperiod, and an additional 12hrs later during the dark phase of the photoperiod, to account for possible differences due to photosynthesis of the *Sargassum* spp.. Five control samples of the ambient seawater from five replicates for each exchange rate were also collected at each sample time. Nitrate and total alkalinity were measured from these samples utilising the methods previously described. pH, temperature and salinity readings were taken for each replicate at each sample time, as previously described. P_{CO_2} concentrations were calculated from temperature, pH, salinity, and total alkalinity data using CO2SYS (Pierrot et al., 2006).

Statistical Analysis

All statistical analyses were conducted using Primer 6 (Primer-E, Plymouth) with PERMANOVA⁺ extension (v.6.1.7) software. Data were analysed using permutational analysis of variance (PERMANOVA, Anderson 2001). PERMANOVA compares the F statistics to a distribution generated by multiple random permutations of the analysed data, thus liberating it from the formal assumptions of ordinary ANOVA (Anderson, 2001). Pair-wise comparisons of untransformed data were generated using Euclidean distance, utilising approximately 9999 permutations of the raw data. A Monte Carlo procedure was used when the number of unique permutations was low. Pair-wise post-hoc tests were performed if PERMANOVA results indicated that there were significant differences between treatments. Repeated measures analyses of TD and WW were conducted for the temperature experiment. Covariates of initial WW and TD were used respectively for analyses of LGR, SGR consumption and FCE. Repeated measures analyses for salinity, pH and P_{CO_2} were conducted for the density and seawater exchange rate experiments.

Permutational analysis of multivariate dispersions (PERMDISP, Anderson, 2006) calculated from Euclidean distances to centroids, utilising approximately 9999 permutations, was conducted on juvenile WW and TD values at week ten for the density and seawater exchange rate experiment. Distance-based linear modelling (DISTLM) using distance based redundancy analysis (dbRDA) (Anderson et al., 2008) was conducted to assess the relative contributions of environmental parameters to variations between treatments in juvenile and adult SGR, LGR and gonad index values respectively in the density and seawater exchange rate experiments. The BEST model selection routine, with R^2 as the selection criterion, based on 9999 permutations, was used to select the environmental parameter that explained the greatest variation amongst treatments out of the total variation explained by all parameters. Environmental parameters tested were standardised experimental mean values for temperature, log transformed pH, nitrate, and P_{CO_2} concentrations.

Results

Experiment 1: The Effects of Temperature on Growth and Survival of *Tripneustes gratilla*

Mean temperatures ($\pm$ SD) in each treatment were 19.63°C ($\pm$ 0.42), 24.1°C ($\pm$ 0.75), 26.6°C ($\pm$ 0.67) and 30.4°C ($\pm$ 0.65) respectively. There was a single mortality in the 29-31°C treatment during the second week of the experiment; there were no mortalities in any other treatments.

The highest cumulative TD growth was 14.4mm to 69.8mm in the 26-28°C treatment (Fig. 4a). Initial TD was not different between treatments (Table 3a; followed by post-hoc pair-wise test, $P < 0.05$, Fig. 4a, not shown), but differentiated after only two weeks. All treatments were statistically different after nine weeks (Table 3a; followed by post-hoc pair-wise test, Fig. 4a,c). The highest TD growth occurred in the 26-28°C treatment, followed by the 23-25°C treatment, and then the 19-21°C treatment, whilst the 29-31°C treatment had the smallest mean TD.

The pattern in TD growth was mirrored by WW growth (Fig. 4b). The highest WW cumulative growth was 1.24g to 88.36g in the 26-28°C treatment. Initial WW was not different between treatments (Table 3b; followed by post-hoc pair-wise test, Fig. 4b, not shown) but differentiated by week two. After nine weeks, WW growth was significantly highest in the 26-28°C treatment, followed by the 23-25°C treatment, and then the 19-21°C treatment, whilst the 29-31°C treatment had the smallest mean WW (Table 3b; followed by post-hoc pair-wise test, Fig. 4b).

At the end of the experiment, all urchins were examined for the development of gonads. 56% of the urchins in the 23-25°C treatment and 67% of the urchins in the 26-28°C treatment had developed gonads, although in all cases this was less than 1.3% of the total WW of the urchins. Urchins in the 19-21°C and 29-31°C treatments did not possess gonads. It was also noted that the colour of the sea urchins' spines and tests were different after nine weeks depending upon temperature treatment (Fig. 4c), but this was not quantified.

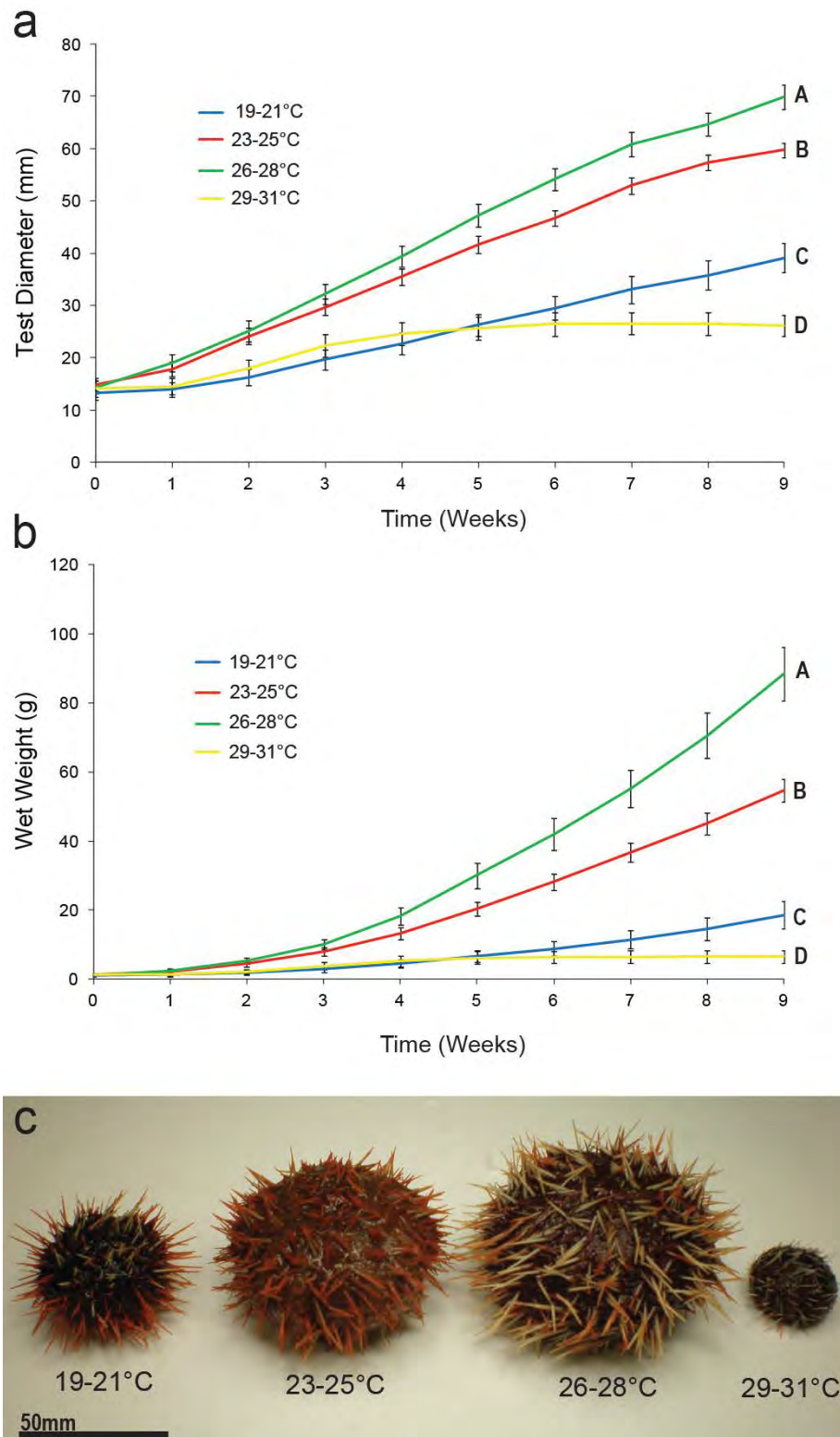


Figure 4: Effect of temperature on the growth of *Tripneustes gratilla* over nine weeks. (a) Cumulative test diameter. (b) Cumulative wet weight. (c) Mean test size and representative spine and test colour variance. Capitalised letters denote statistical differences between treatments at nine weeks (Table 3a,b respectively; followed by post-hoc pair-wise test, $P < 0.05$). Data are means $\pm$ S.E.

Table 3: Repeated measures PERMANOVA analyses examining the effects of temperature on (a) cumulative test diameter and (b) cumulative wet weight growth of *Tripneustes gratilla* over nine weeks.

a						b					
Source	df	MS	F	P	Unique Permutations	Source	df	MS	F	P	Unique Permutations
Temperature	3	84.378	28.124	0.0001	9960	Temperature	3	14623	33.814	0.0001	9959
Time	9	54.235	2015.7	0.0001	9944	Time	9	7255.1	302.59	0.0001	9939
Container (Temperature)	31	3.0003	111.51	0.0001	9890	Container (Temperature)	31	432.46	18.037	0.0001	9890
Temperature x Time	27	4.024	149.55	0.0001	9895	Temperature x Time	27	1491.6	62.211	0.0001	9911
Residual	279	0.026906				Residual	279	23.976			
Total	349					Total	349				

Significant differences ($P < 0.05$) are in bold; df, degrees of freedom, MS, mean square

Experiment 2: The Effects of Density and Seawater Exchange Rate on Juvenile Growth and Survival

Effect of Density and Seawater Exchange Rate on Juvenile Growth and Survival

Mortalities were observed in five of the nine treatments within the first two weeks of the experiment; there were no further mortalities after this time. Survival was $>90\%$ in all treatments (Fig. 5), and there was no significant difference in survival between seawater exchange rates or culture densities after ten weeks (Table 4a). There was high mortality ($\sim 47\%$) in one replicate of the Low exchange rate, High density treatment, with deceased urchins displaying symptoms of disease. This replicate was excluded from further statistical analysis.

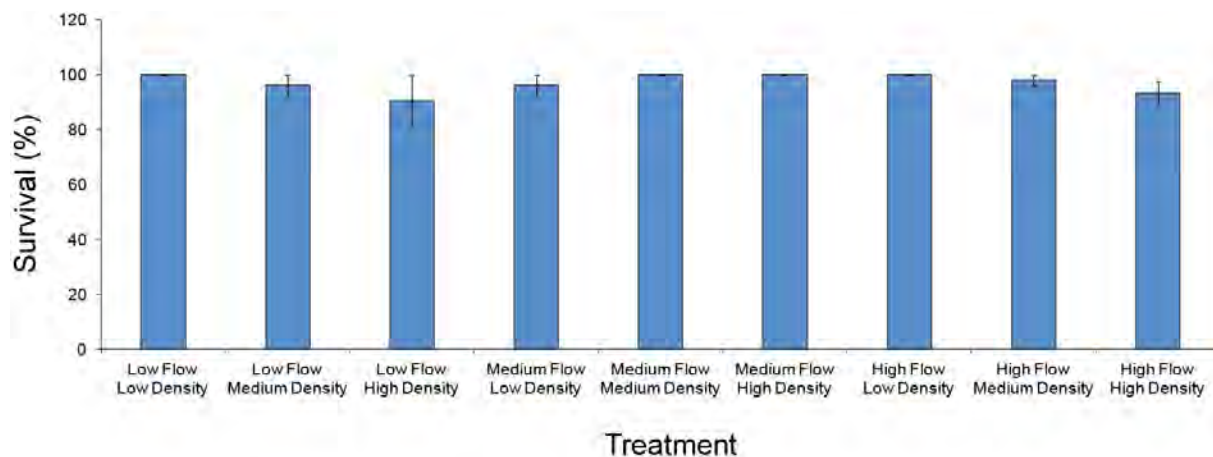


Figure 5: Effect of density and seawater exchange rate (Flow) on survival of *Tripneustes gratilla* over ten weeks in a flow-through seawater system. There were no statistical differences between treatments (Table 4a). Data are means $\pm$ S. E.

Table 4: PERMANOVA analyses examining the effects of density and seawater exchange rate on (a) survival, (b) Specific Growth Rate and (c) Linear Growth Rate of juvenile *Tripneustes gratilla* after ten weeks

a					
Source	df	MS	F	P	Unique Permutations
Flow	2	36.296	0.46372	0.6682	443
Density	2	68.889	0.88001	0.4559	440
Flow x Density	4	62.963	0.80442	0.5815	1083
Residual	36	78.272			
Total	44				

b					
Source	df	MS	F	P	Unique Permutations
Flow	2	47.68	136.85	0.0001	9954
Density	2	99.212	284.76	0.0001	9946
Initial Weight	1	0.45702	1.3117	0.2636	9851
Flow x Density	4	1.9619	5.6312	0.0017	9946
Residual	34	0.3484			
Total	43				

c					
Source	df	MS	F	P	Unique Permutations
Flow	2	2.8303	119.89	0.0001	9955
Density	2	5.3468	226.48	0.0001	9958
Initial Diameter	1	0.0042602	0.18045	0.6809	9827
Flow x Density	4	0.086503	3.6641	0.0124	9950
Residual	34	0.023609			
Total	43				

Significant differences ($P < 0.05$) are in bold; df, degrees of freedom, MS, mean square

The highest mean cumulative WW growth after ten weeks was 4.06g to 100.4g in the High exchange rate, Low density treatment (Fig. 6a, 7). SGR was significantly affected by both density and seawater exchange rate (Table 4b, followed by post-hoc pair-wise test, Fig. 6b). In general, increased density reduced SGR, whilst increased exchange rate had the opposite effect. There was an interaction between density and exchange rate (Table 4b); decreasing exchange rate increased the effect of density on SGR (Fig. 6b). There was no significant difference in variation of WW within treatments at week ten for different exchange rates (PERMDISP, $F_{2, 423} = 0.22626$, $P = 0.8182$) or densities (PERMDISP, $F_{2, 423} = 0.087745$, $P = 0.9158$).

The highest mean cumulative TD growth after ten weeks was 21.7mm to 67.6mm in the High exchange rate, Low density treatment (Fig. 6c, 7). LGR followed the same pattern as for SGR. Both density and exchange rate had a significant effect on LGR (Table 4c, followed by post-hoc pair-wise test, Fig. 6d). Increased density reduced LGR, whilst increased exchange rate increased LGR. There was also an interaction between the two factors that mirrored the interaction seen in SGR (Table 4c, Fig. 6d); decreasing exchange rate increased the effect of density on LGR. There were significant differences in the variation of TD within treatments at week ten between both seawater exchange rate (PERMDISP, $F_{2, 423} = 4.3389$, $P = 0.021$, followed by post-hoc pair-wise test) and density (PERMDISP, $F_{2, 423} = 4.0267$, $P = 0.022$, followed by post-hoc pair-wise test). Medium and High seawater exchange rates had significantly increased TD variation compared to Low exchange rates. Variation in TD was significantly greater in High density compared to Low density treatments.

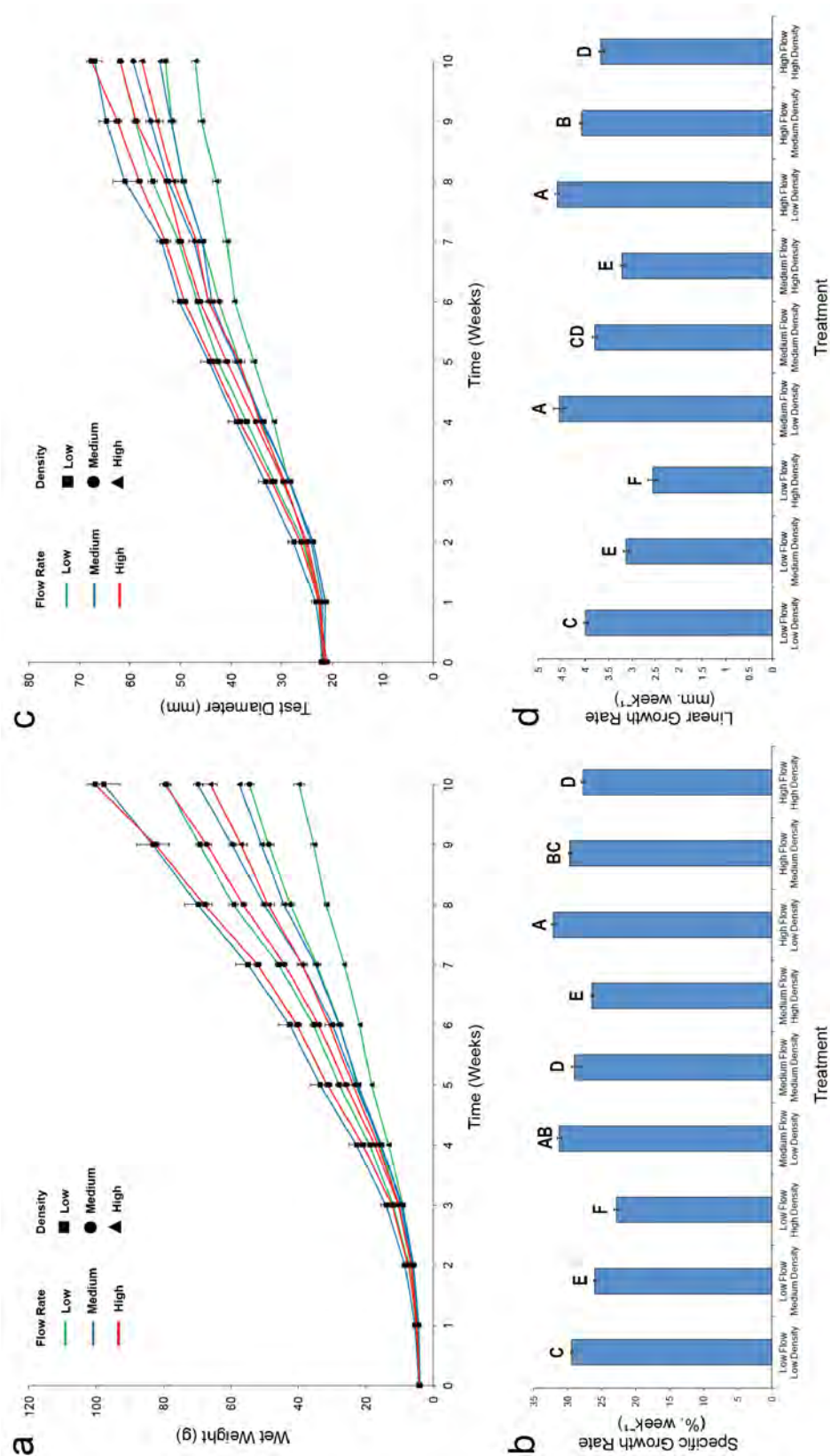


Figure 6: Effects of density and seawater exchange rate (Flow) on the growth of *Tripneustes gratilla* over ten weeks in a flow-through seawater system. (a) Cumulative wet weight. (b) SGR (Specific Growth Rate). (c) Cumulative test diameter. (d) LGR (Linear Growth Rate). Capitalised letters denote statistical differences between treatments (Table 4b,c respectively; followed by post-hoc pair-wise test, $P < 0.05$). Data are means $\pm$ S.E.

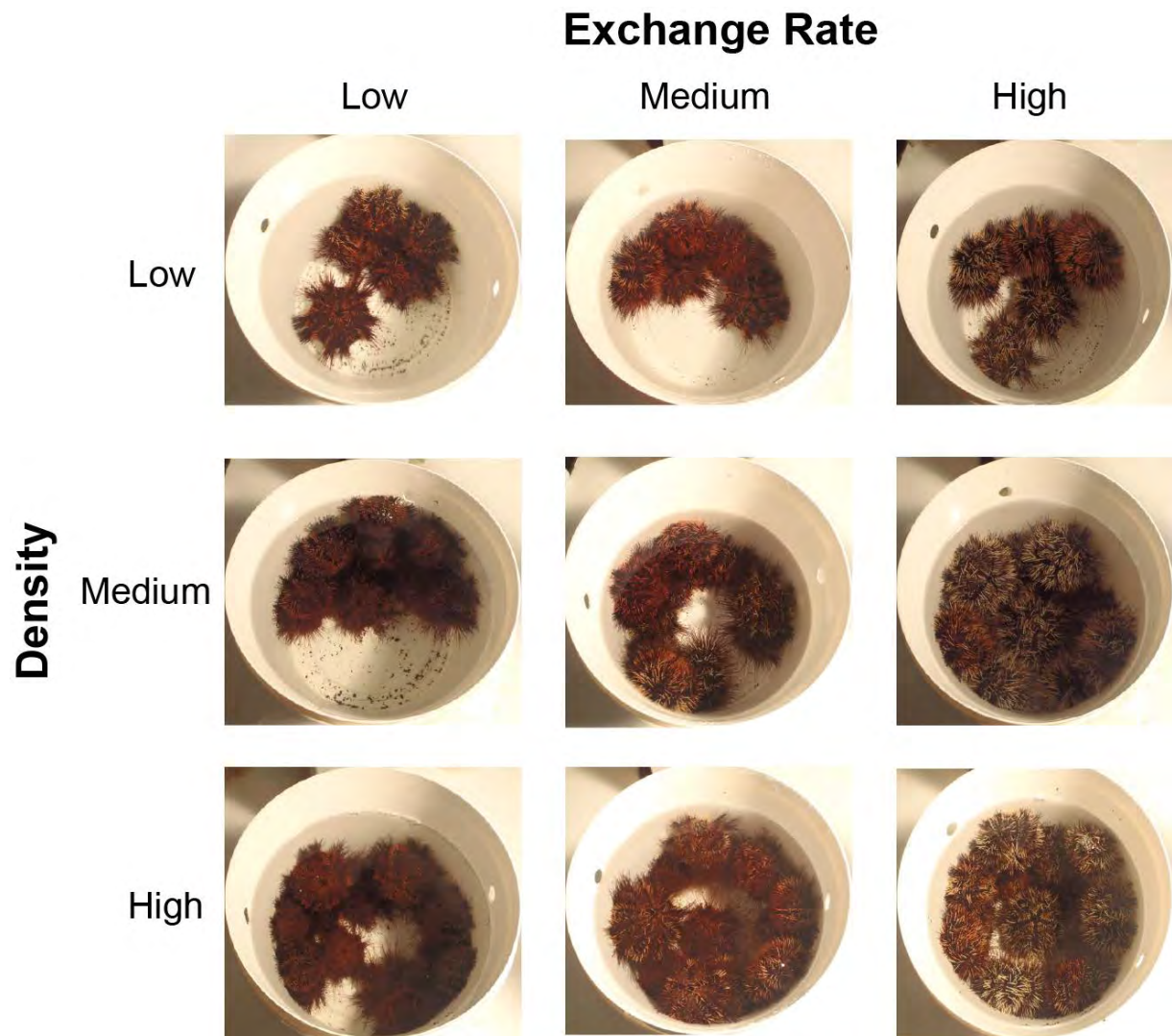


Figure 7: Representative size and densities of juvenile *Tripneustes gratilla* after ten weeks in a flow-through culture system under different density and seawater exchange rate regimes.

Effects of Density and Seawater Exchange Rate on Juvenile Consumption and FCE

Juvenile sea urchins consumed 7-9% of their body mass per day (Fig. 8a). Consumption was significantly affected by both seawater exchange rate and density (Table 5a, followed by pair-wise post-hoc test, Fig. 8a). Consumption was highest in the Low and Medium densities, significantly higher than consumption at the High density. Consumption significantly increased with each increase in seawater exchange rate.

FCE of juvenile sea urchins was greater than 50% (Fig. 8b). FCE was significantly affected by both density and seawater exchange rate (Table 5b, followed by pair-wise post-hoc test, Fig. 8b). FCE was highest in the Low and Medium densities, significantly higher than FCE at the High density. FCE was highest in the Medium and High exchange rates, significantly higher than FCE at the Low exchange rate.

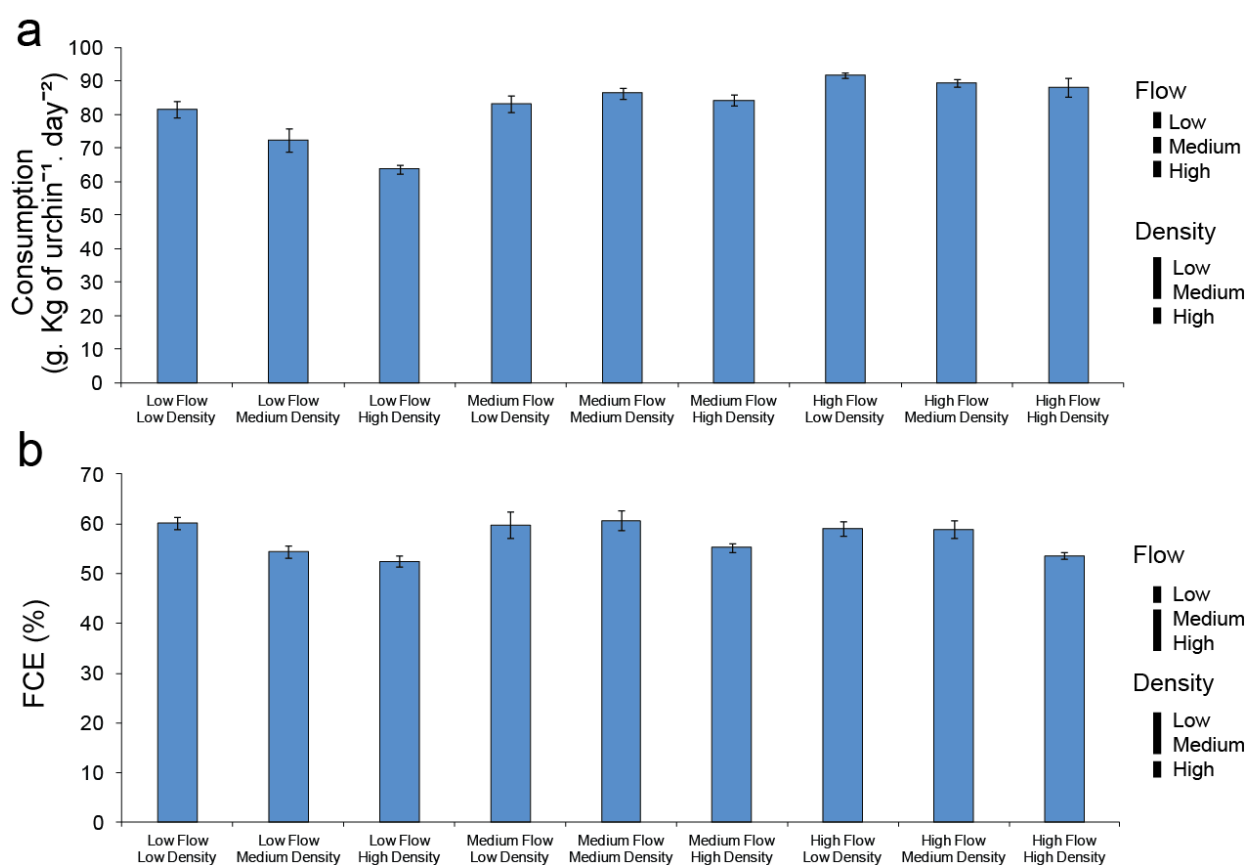


Figure 8: Effects of density and seawater exchange rate (Flow) on (a) consumption and (b) food conversion efficiency (FCE) of juvenile *Tripneustes gratilla* over 20 days in a flow-through seawater system. Bars denote statistical differences between treatments (Table 5a,b respectively; followed by post-hoc pair-wise test, $P < 0.05$). Data are means $\pm$ S.E.

Table 5: PERMANOVA analyses examining the effects of density and seawater exchange rate on (a) consumption and (b) food conversion efficiency (FCE) of *Tripneustes gratilla* over 20 days in a flow-through seawater system

a						b					
Source	df	MS	F	P	Unique Permutations	Source	df	MS	F	P	Unique Permutations
Flow (Fl)	2	1.0819*10 ⁻³	79.4	0.0001	9959	Flow	2	83.756	10.889	0.0003	9956
Density (De)	2	1.4269*10 ⁻³	10.472	0.001	9933	Density	2	154.28	20.057	0.0001	9942
Initial Weight (Co)	1	8.6058*10 ⁻⁵	6.3158	0.0213	9843	Initial Weight	1	19.398	2.5218	0.1243	9831
Fl x Co	2	3.3597*10 ⁻⁴	24.657	0.0001	9961	Flow x Density	4	11.353	1.4759	0.2236	9954
De x Co	2	1.8287*10 ⁻⁵	0.13421	0.8748	9947	Residual	33	7.6921			
Flow x Density	4	8.2686*10 ⁻⁵	0.60684	0.6577	9954	Total	42				
Fl x De x Co	4	1.0776*10 ⁻⁵	0.79086	0.539	9960						
Residual	26	1.3626*10 ⁻⁵									
Total	43										

Significant differences ($P < 0.05$) are in bold; df, degrees of freedom, MS, mean square

Effects of Density and Seawater Exchange Rate on Environmental Parameters

Temperatures were maintained within the optimal temperature range for growth of *T. gratilla* (26–28°C, see above). Temperatures were significantly different between seawater exchange rates, but not densities (Table 6a, followed by pair-wise post-hoc test, $P < 0.05$, Fig. 9a). Temperature was highest in the High exchange treatments, decreasing with decreasing exchange rate.

Nitrate concentrations were lower in all treatments compared to mean concentration (0.44mg.L⁻¹) detected in the seawater control samples (Fig. 9b). Nitrate concentration was significantly different between seawater exchange rates (Table 6b, followed by pair-wise post-hoc test, Fig. 9b). Nitrate concentration significantly increased with each increase in exchange rate.

TAN and nitrite were not detected (<0.00mg.L⁻¹), except during week five for a period of 72hrs, at a maximum of 0.06mg.L⁻¹ TAN and 0.02mg.L⁻¹ nitrite in the intake seawater. This was associated with a heavy rainfall event.

Salinity dropped from ~36ppt to as low as ~28ppt due to rainfall events during the first six weeks of the experiment (Fig. 9c), but increased thereafter. Salinity was significantly different between exchange rates (Table 6c, followed by pair-wise post-hoc tests, $P < 0.01$, Fig. 9c, not shown). Salinity was always significantly higher in the Low exchange rate treatments than in the Medium and High exchange rate treatments.

pH decreased in all treatments over time (Fig. 9d). pH was significantly different between both exchange rates and densities (Table 6d, followed by pair-wise post-hoc tests, $P < 0.05$ Fig. 9d, not shown). pH was significantly lower in the Low exchange in comparison to the Medium or High exchange rate treatments. pH was significantly lower in the High density than in the Low or Medium densities, which were significantly different, except at week ten.

P_{CO_2} increased in all treatments over time (Fig. 9e). P_{CO_2} was significantly different between both exchange rates and densities (Table 6e, followed by pair-wise post-hoc tests, $P < 0.05$, Fig. 9e, not shown). P_{CO_2} was significantly lower in the High exchange than in the Low or Medium exchange rate treatments over the ten weeks. P_{CO_2} was significantly lower in the Low density than in the Medium or High densities, which were not significantly different over the ten weeks. There was also a significant interaction between density and exchange rate (Table 6e). Decreased exchange rate increased the effect of density on P_{CO_2} (Fig. 9e, not shown).

Mean total alkalinity was lower in all treatments compared to the ambient seawater supplied to the system (Fig. 9f). Total alkalinity was significantly affected by both exchange rate and density (Table 6f, followed by pair-wise post-hoc tests, Fig. 9f). Total alkalinity was highest in the High exchange rate treatments, decreasing with decreasing exchange rate. Total alkalinity was significantly higher in the Low density than in the Medium or High densities, which did not differ.

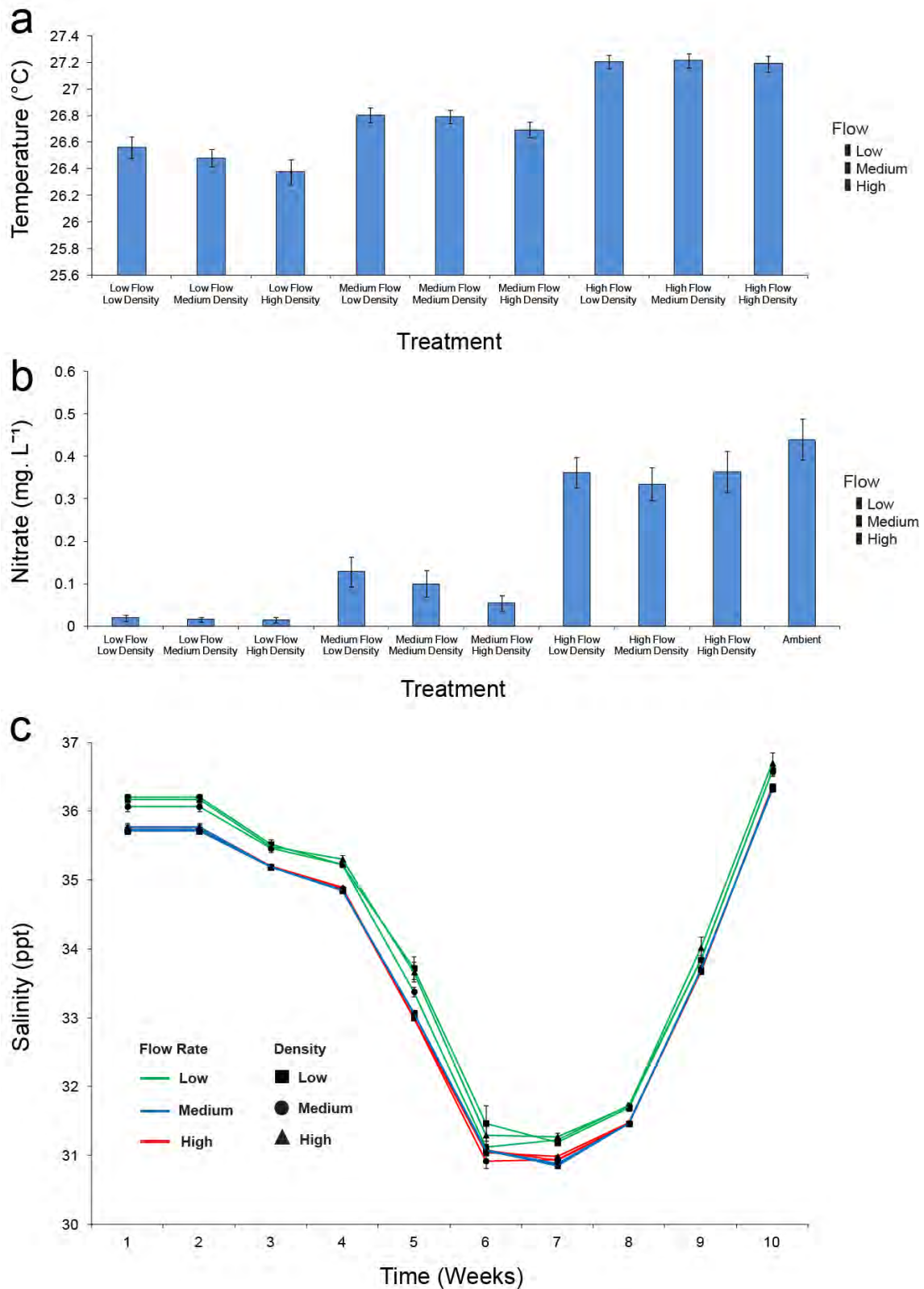


Figure 9: The effects of seawater exchange rate (Flow) and density of *Tripneustes gratilla* on environmental parameters in a flow-through seawater system over ten weeks. (a) Temperature. (b) Nitrate concentration. (c) Salinity. Bars denote statistical differences between treatments (Table 6a,b respectively; followed by post-hoc pair-wise test, $P < 0.05$). Ambient = Control seawater samples. Data are means $\pm$ S.E.

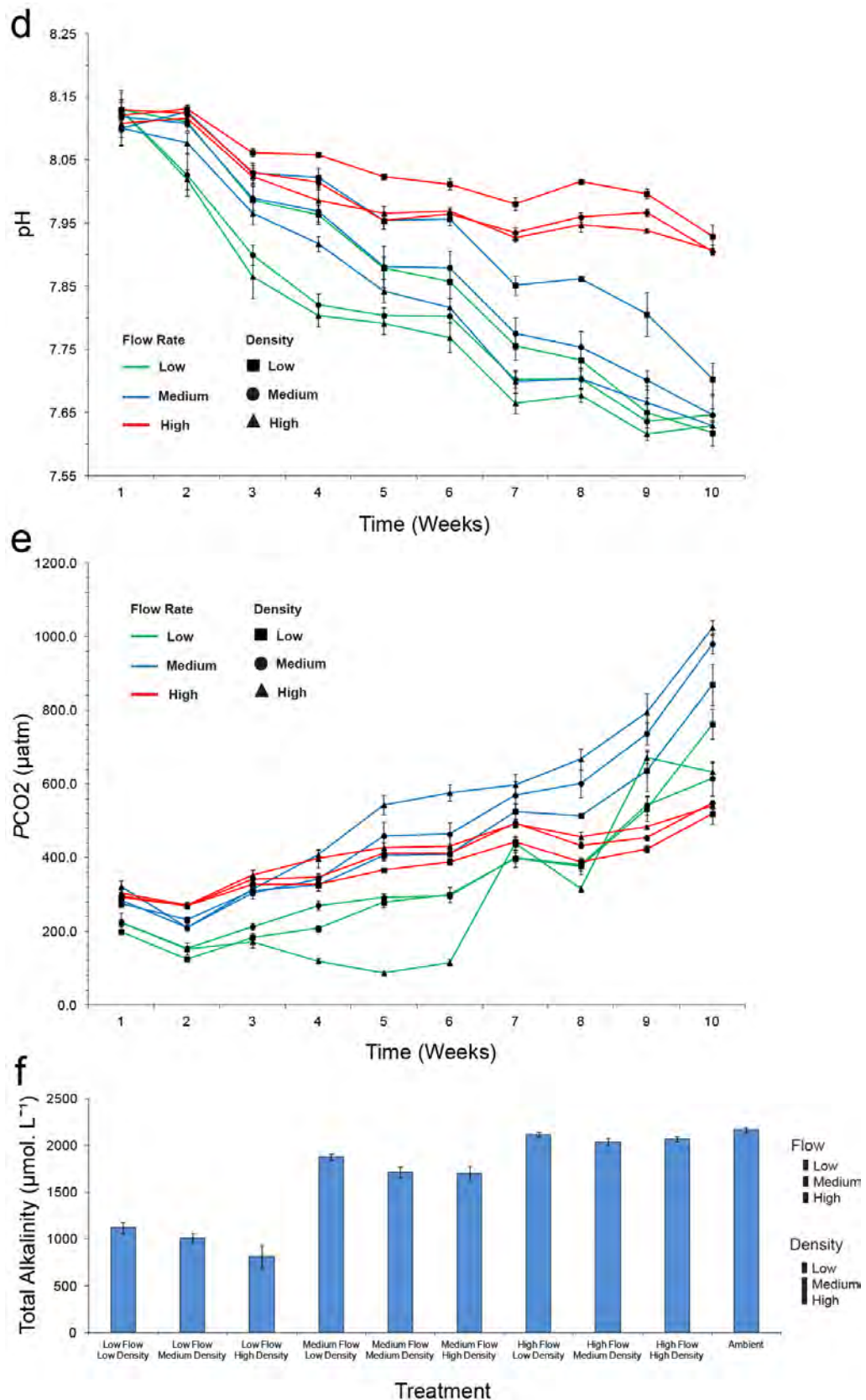


Figure 9: The effects of seawater exchange rate (Flow) and density of *Tripteneustes gratilla* on environmental parameters in a flow-through seawater system over ten weeks. (d) pH. (e) Carbon dioxide concentration (P_{CO_2}). (f) Total Alkalinity. Bars denote statistical differences between treatments (Table 6f; followed by post-hoc pair-wise test, $P < 0.05$). Ambient = Control seawater samples. Data are means $\pm$ S.E.

Table 6: PERMANOVA analyses examining the effects of seawater exchange rate (Flow) and density of *Tripneustes gratilla* on (a) temperature, (b) nitrate concentration, (c) salinity, (d) pH, (e) carbon dioxide and (f) total alkalinity concentrations in a flow-through seawater system over ten weeks.

a						
Source	df	MS	F	P	Unique Permutations	
Flow	2	4.0074	96.028	0.0001	9952	
Density	2	0.068268	1.6359	0.1988	9945	
Flow x Density	4	0.025606	0.61359	0.6578	9941	
Residual	81	0.041731				
Total	89					

b						
Source	df	MS	F	P	Unique Permutations	
Flow	2	0.93046	90.916	0.0001	9942	
Density	2	0.0054927	0.5812	0.5812	9945	
Flow x Density	4	0.005464	0.7031	0.7031	9936	
Residual	90	0.010234				
Total	99					

c						
Source	df	MS	F	P	Unique Permutations	
Flow	2	5.3897	283.57	0.0001	9945	
Density	2	0.088128	4.6366	0.0156	9953	
Flow x Density	4	0.04310	2.2678	0.082	9958	
Time	9	191.57	14064	0.0001	9932	
Container (Flow, Density)	35	0.019007	1.3954	0.0781	9891	
Flow x Time	18	0.056734	4.1651	0.0001	9910	
Density x Time	18	0.015379	1.129	0.3247	9917	
Flow x Density x Time	36	0.010947	0.80369	0.7855	9891	
Residual	315	0.013621				
Total	439					

d						
Source	df	MS	F	P	Unique Permutations	
Flow	2	1.3066	254.64	0.0001	9937	
Density	2	0.22325	43.508	0.0001	9947	
Flow x Density	4	0.011721	2.2842	0.0898	9953	
Time	9	0.68894	722.33	0.0001	9937	
Container (Flow, Density)	35	0.0051312	5.3799	0.0001	9882	
Flow x Time	18	0.041626	43.643	0.0001	9907	
Density x Time	18	0.0027223	2.8542	0.0002	9912	
Flow x Density x Time	36	0.0021921	2.2984	0.0002	9886	
Residual	315	0.0095378				
Total	439					

e						
Source	df	MS	F	P	Unique Permutations	
Flow	2	1.1097*10 ⁶	168.56	0.0001	9945	
Density	2	39141	5.9455	0.0068	9956	
Flow x Density	4	59504	9.0385	0.0001	9952	
Time	9	1.1027*10 ⁶	712.21	0.0001	9920	
Container (Flow, Density)	35	6583.4	4.252	0.0001	9887	
Flow x Time	18	1.0248*10 ⁵	66.191	0.0001	9922	
Density x Time	18	5882.2	3.7992	0.0001	9917	
Flow x Density x Time	36	10660	6.8848	0.0001	9892	
Residual	315	1548.3				
Total	439					

f						
Source	df	MS	F	P	Unique Permutations	
Flow	2	9.5221*10 ⁻⁶	262.71	0.0001	9946	
Density	2	2.4291*10 ⁻⁵	6.7018	0.0022	9953	
Flow x Density	4	53574	1.4781	0.2103	9939	
Residual	90	36246				
Total	99					

c,d,e are Repeated Measures PERMANOVA analyses.

Significant differences (P<0.05) are in bold; df, degrees of freedom, MS, mean square

Effects of Environmental Parameters on Juvenile Growth

DISTLM analyses indicated that all 100% of all variation in juvenile SGR and LGR could be explained by the first PCO axis; all variation could be attributed to differences between treatments. Temperature, pH, nitrate, and P_{CO_2} cumulatively accounted for 68.62% and 64.67% of the total variation for SGR and LGR respectively. The individual parameter that accounted for the greatest total variation in both SGR and LGR was pH (53.22% and 55.78% respectively).

Experiment 3: The Effects of Density and Seawater Exchange Rate on Adult Growth and Survival

Effect of Density and Seawater Exchange Rate on Adult Growth and Survival

There were no mortalities in any of the treatments over the six weeks. The highest mean cumulative WW growth was 105.8g to 244.7g in the Low exchange rate, Low density treatment (Fig. 10a, 11). SGR was significantly affected by both density and exchange rate (Table 7a, followed by post-hoc pair-wise test, Fig. 10b). Increases in density significantly reduced SGR. SGR was significantly higher in the High and Medium exchange rate treatments compared to the Low exchange rate treatments. There was a significant interaction between density and exchange rate (Table 7a); decreasing exchange rate increased the effect of density on SGR.

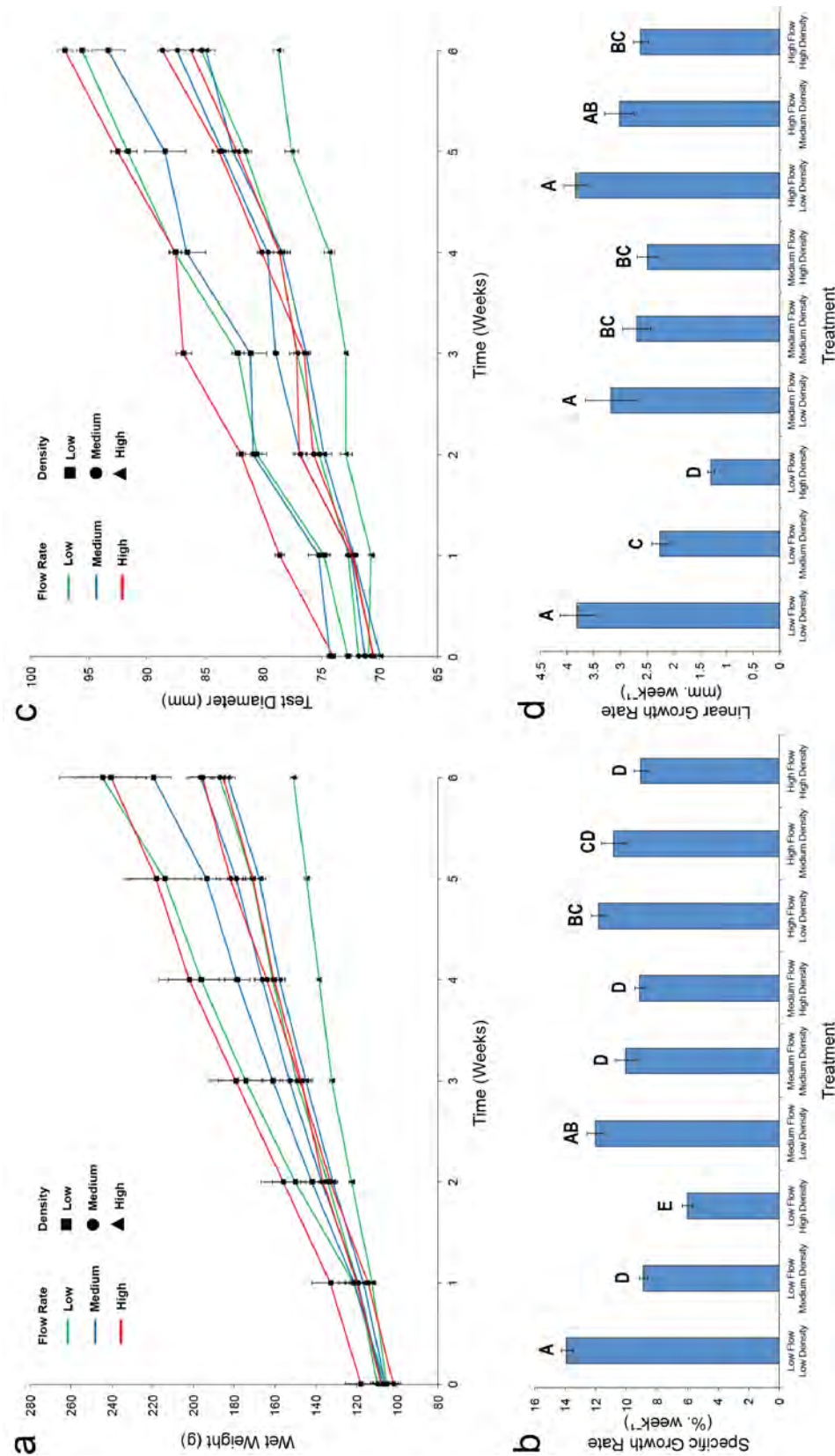


Figure 10: Effects of density and seawater exchange rate on growth of adult *Tripneustes gratilla* after six weeks in a flow-through seawater system. a) Cumulative wet weight. (b) SGR (Specific Growth Rate). (c) Cumulative test diameter. (d) LGR (Linear Growth Rate). Capitalised letters denote statistical differences between treatments (Table 7, followed by post-hoc pair-wise test, $P < 0.05$). Data are means $\pm$ S.E.

The highest mean cumulative TD growth after ten weeks was 74.0mm to 97.0mm (Fig. 10c). LGR followed the same pattern as for SGR. Both density and exchange rate had a significant effect on LGR (Table 7b, followed by post-hoc pair-wise test, Fig. 10d). Increased density reduced LGR, whilst increased exchange rate increased LGR. There was also an interaction between exchange rate and density (Table 7b). Decreasing exchange rate increased the effect of density on LGR (Fig. 10d).

Table 7: PERMANOVA analyses examining the effects of density and seawater exchange rate on (a) Specific Growth Rate, (b) Linear Growth Rate and (c) Gonad Index of adult *Tripneustes gratilla* after six weeks.

a					
Source	df	MS	F	P	Unique Permutations
Flow	2	4.1583	3.4337	0.0419	9941
Density	2	81.843	67.58	0.0001	9931
Initial Weight	1	0.84413	0.69703	0.4148	9829
Flow x Density	4	10.043	8.2931	0.0002	9955
Residual	35	1.211			
Total	44				

b					
Source	df	MS	F	P	Unique Permutations
Flow	2	1.9042	5.8922	0.0051	9959
Density	2	9.3309	28.872	0.0001	9936
Initial Diameter	1	0.30619	0.94742	0.3368	9843
Flow x Density	4	0.89786	2.7782	0.0393	9946
Residual	35	0.32318			
Total	44				

c					
Source	df	MS	F	P	Unique Permutations
Flow (Fl)	2	14.584	14.577	0.0001	9946
Density (De)	2	18.83	18.821	0.0001	9960
Final Weight (Co)	1	5.3063	5.3036	0.0334	9838
Fl x Co	2	1.4544	1.4537	0.2587	9959
De x Co	2	0.1116	0.11154	0.8936	9964
Flow x Density	4	0.73553	0.73516	0.5713	9956
Fl x De x Co	4	0.67466	0.67432	0.5968	9947
Residual	27	1.0005			
Total	44				

Significant differences ($P < 0.05$) are in bold; df, degrees of freedom, MS, mean square

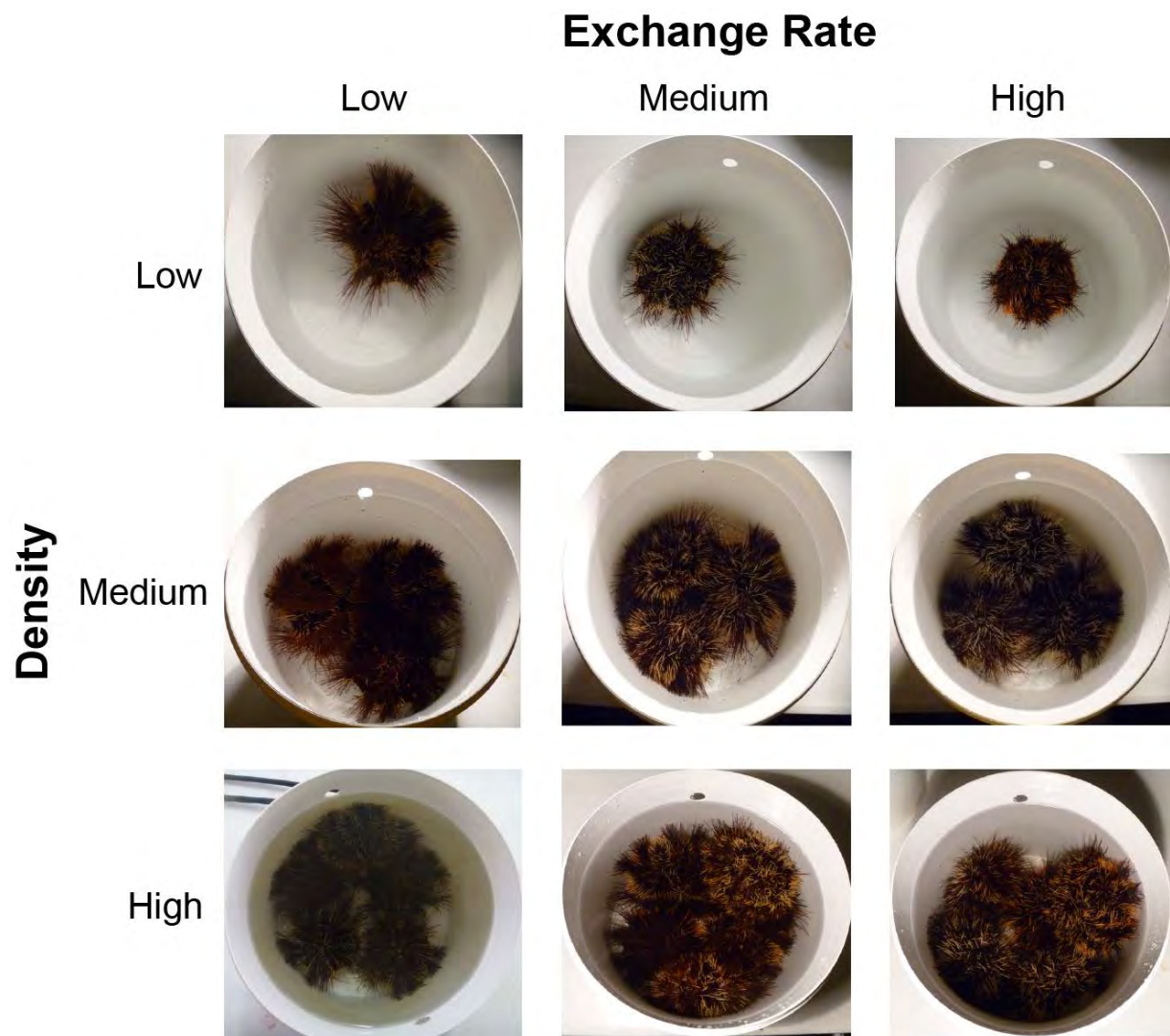


Figure 11: Representative size and densities of adult *Tripneustes gratilla* after six weeks in a flow-through culture system under different density and seawater exchange rate regimes.

Effects of Density and Seawater Exchange Rate on Adult Gonad Production

All *T. gratilla* developed higher GIs over the six weeks (Fig. 12) compared to initial controls ($1.43\% \pm 0.21\text{S.E.}$). GI was significantly affected by both exchange rate and density (Table 7c, followed by pair-wise post-hoc test, Fig. 12). Increases in density significantly decreased GI. GI was significantly lower at the Low exchange rate compared to the Medium and High exchange rates, which did not differ.

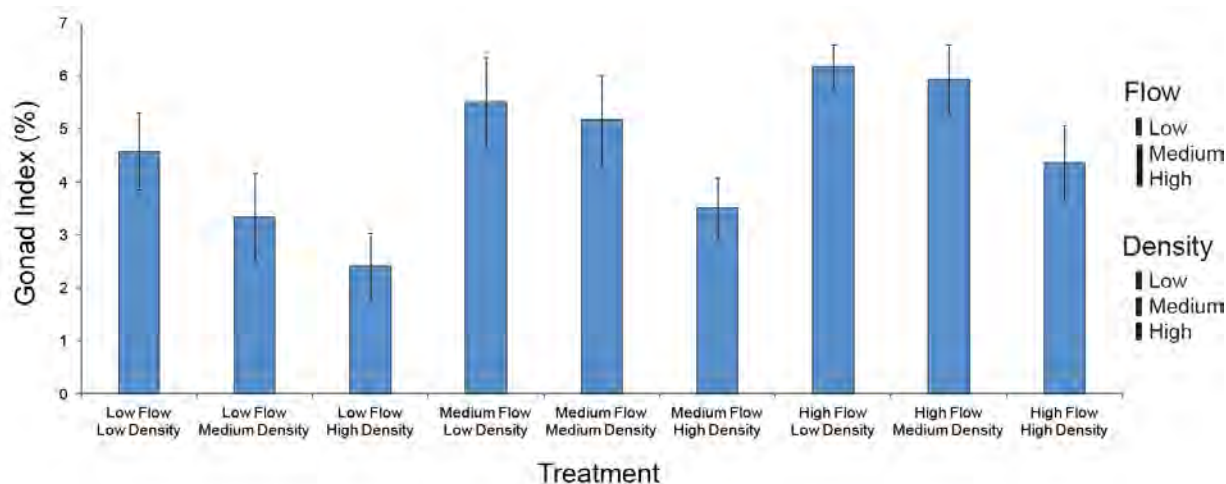


Figure 12: Effect of density and seawater exchange rate (Flow) on gonad indices of *Tripneustes gratilla* after six weeks in a flow-through seawater system. Bars denote statistical differences between treatments (Table 7; followed by post-hoc pair-wise test, $P < 0.05$). Data are means $\pm$ S.E.

Effects of Density and Seawater Exchange Rate on Adult Consumption and FCE

Adult sea urchins consumed 2-6% of their body mass per day (Fig. 13a). Consumption was significantly affected by both seawater exchange rate and density (Table 8a, followed by pair-wise post-hoc test, Fig. 13a). Consumption was highest in the Low and Medium densities, significantly higher than the High density treatments. Consumption was significantly lower in the Low exchange rate treatments compared to the Medium and High exchange rates treatments, which did not differ.

Adult FCE was 20-40% in all treatments. PERMANOVA analysis was unable to detect differences in FCE between treatments, likely due to the high variability of FCE within treatments (Table 8b, Fig. 13b).

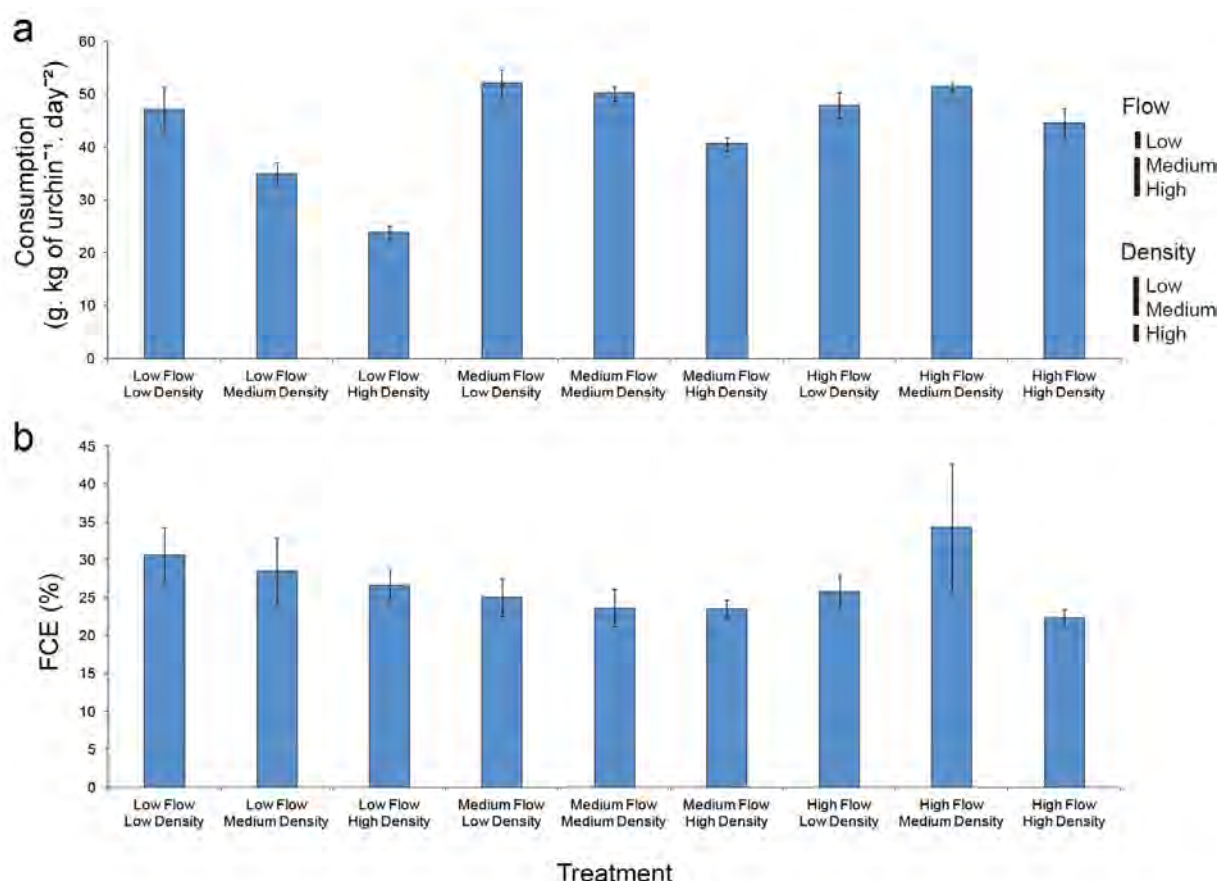


Figure 13: Effects of density and seawater exchange rate (Flow) on (a) consumption and (b) food conversion efficiency (FCE) of adult *Tripneustes gratilla* over seven days in a flow-through seawater system. Bars denote statistical differences in consumption between treatments (Table 8a; followed by post-hoc pair-wise test, $P < 0.05$). There were no significant differences between treatments for FCE (Table 8b). Data are means $\pm$ S.E.

Table 8: PERMANOVA analyses examining the effects of density and seawater exchange rate (Flow) on (a) consumption and (b) food conversion efficiency (FCE) of adult *Tripneustes gratilla* over seven days in a flow-through seawater system

a						b					
Source	df	MS	F	P	Unique Permutations	Source	df	MS	F	P	Unique Permutations
Flow (Fl)	2	706.3	39.014	0.0001	9957	Flow	2	104.21	2.0685	0.1411	9940
Density (De)	2	221.57	12.244	0.0003	9955	Density	2	69.236	1.3744	0.2697	9957
Initial Size (Co)	1	1008.3	55.719	0.0001	9823	Initial Size	1	217.3	4.3134	0.0448	9851
Fl x Co	2	421.95	23.316	0.0001	9958	Flow x Density	4	47.949	0.95181	0.4536	9944
De x Co	2	65.605	3.6252	0.0386	9948	Residual	33	50.377			
Flow x Density	4	0.87665	0.048442	0.9947	9950	Total	42				
Fl x De x Co	4	11.132	0.61515	0.6538	9950						
Residual	27	18.097									
Total	44										

Significant differences ($P < 0.05$) are in bold; df, degrees of freedom, MS, mean square

Effects of Density and Seawater Exchange Rate on Environmental Parameters

Temperatures were maintained within the optimal temperature range for growth of *T. gratilla* (26-28°C, Fig. 14a). Temperatures were significantly different between seawater exchange rates, but not densities (Table 9a, followed by pair-wise post-hoc test, $P < 0.05$, Fig. 14a). Temperature was highest in the High exchange treatments, decreasing with decreasing exchange rate.

Nitrate concentrations were lower in all treatments compared to mean concentration (0.33 mg.L^{-1}) detected in the seawater control samples (Fig. 14b). Nitrate concentration was significantly different between seawater exchange rates (Table 9b, followed by pair-wise post-hoc test, Fig. 14b). Nitrate concentration significantly increased with each increase in exchange rate.

TAN and nitrite were not detected ($< 0.00 \text{ mg.L}^{-1}$) at any time during the experiment.

pH decreased in some of the treatments over the six weeks (Fig. 14c). pH was significantly different between both seawater exchange rates and densities over the six weeks (Table 9c, followed by pair-wise post-hoc tests, Fig. 14c, not shown). pH was significantly decreased by decreases in exchange rate, and increases in density. There was also a significant interaction between exchange rate and density; decreased exchange rates increased the effect of density.

P_{CO_2} concentration increased in some treatments over the six weeks (Fig. 14d). P_{CO_2} was significantly different between both exchange rates and densities (Table 9e, followed by pair-wise post-hoc tests, Fig. 14d, not shown). P_{CO_2} significantly increased with each increase in density and with each decrease in exchange rate. There was also a significant interaction between density and exchange rate (Table 9d). In general, decreased exchange increased the effect of density on P_{CO_2} concentration, and the effect of this interaction strengthened over time (Table 9e, Fig. 14d).

Total alkalinity was lower in all treatments compared to the ambient seawater supplied to the system. Total alkalinity was significantly affected by both exchange rate and density (Table 9e, followed by pair-wise post-hoc tests, Fig. 14e). Total alkalinity was highest in the High exchange rate treatments, decreasing with decreasing exchange rate. Total alkalinity was significantly higher in the Low density than in the Medium or High densities, which did not differ. There was also a significant interaction between density and exchange rate (Table 9e). Decreasing exchange rate increased the effect of density on total alkalinity concentration.

Salinity did not vary substantially over the six weeks of the experiment (Fig. 14f). Salinity was significantly different between seawater exchange rates, but not densities (Table 9f, followed by pair-wise post-hoc tests, Fig. 14f). Salinity was significantly higher in the Low exchange rate treatment than in the Medium and High exchange rate treatments.

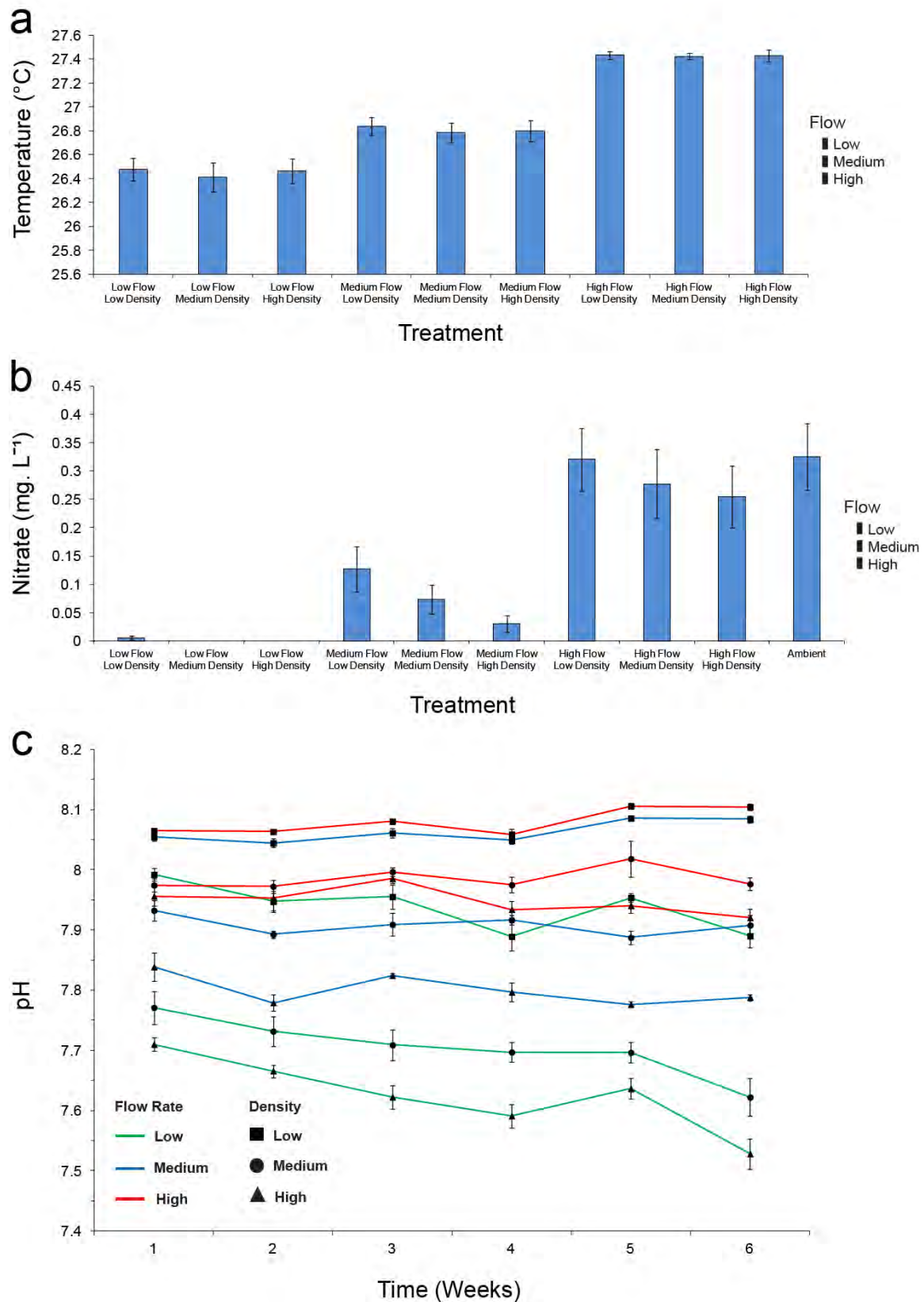


Figure 14: The effects of seawater exchange rate (Flow) and density of adult *Tripneustes gratilla* on environmental parameters in a flow-through seawater system over six weeks. (a) Temperature. (b) Nitrate concentration. (c) pH. Bars denote statistical differences between treatments (Table 9a,b respectively; followed by post-hoc pair-wise test, $P < 0.05$). Ambient = Control seawater samples. Data are means $\pm$ S.E.

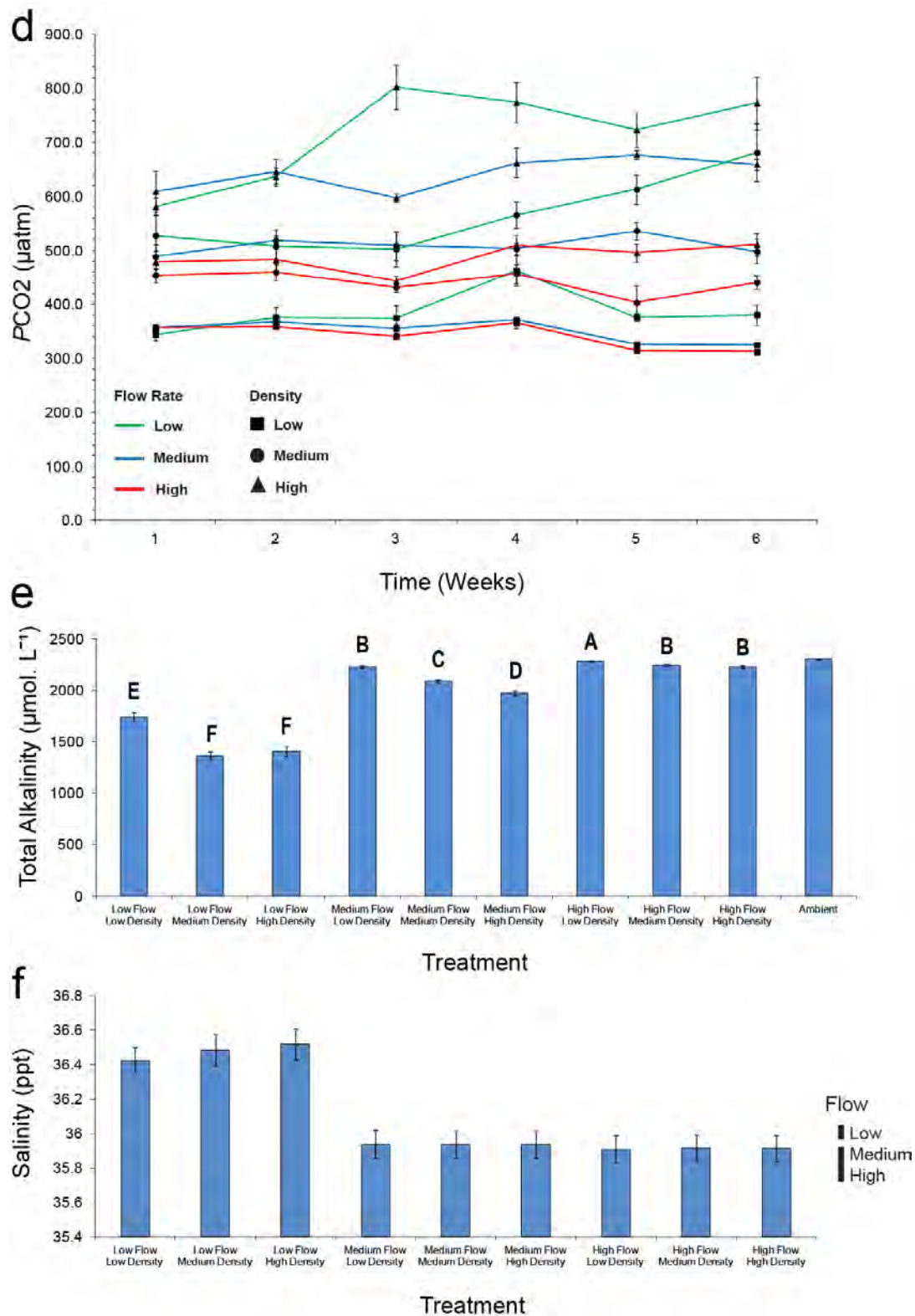


Figure 14: The effects of seawater exchange rate (Flow) and density of adult *Tripneustes gratilla* on environmental parameters in a flow-through seawater system over six weeks. (d) Carbon dioxide concentration (P_{CO_2}). (e) Total Alkalinity. (f) Salinity. Bars and capitalised letters denote statistical differences between treatments (Table 9e,f respectively; followed by post-hoc pair-wise test, $P < 0.05$). Ambient = Control seawater samples. Data are means $\pm$ S.E.

Table 9: PERMANOVA analyses examining the effects of seawater exchange rate (Flow) and density of adult *Tripneustes gratilla* on (a) temperature, (b) nitrate concentration, (c) pH, (d) carbon dioxide, (e) total alkalinity and (f) salinity in a flow-through seawater system over six weeks

a						
Source	df	MS	F	P	Unique Permutations	
Flow	2	4.3878	111.28	0.0001	9952	
Density	2	0.0078285	0.19855	0.8174	9952	
Flow x Density	4	0.001912	0.048492	0.9955	9961	
Residual	45	0.039429				
Total	53					

b						
Source	df	MS	F	P	Unique Permutations	
Flow	2	0.38519	41.043	0.0001	9961	
Density	2	0.014213	1.5144	0.2349	9948	
Flow x Density	4	0.003215	0.34257	0.8494	9965	
Residual	50	0.0093851				
Total	59					

c						
Source	df	MS	F	P	Unique Permutations	
Flow	2	1.4465	404.29	0.0001	9954	
Density	2	1.3043	364.55	0.0001	9965	
Flow x Density	4	0.074161	20.728	0.0001	9960	
Time	5	0.015942	23.3	0.0001	9942	
Container (Flow, Density)	36	0.0035779	5.2295	0.0001	9897	
Flow x Time	10	0.012726	18.601	0.0001	9938	
Density x Time	10	0.0039078	5.7116	0.0002	9935	
Flow x Density x Time	20	8.1479*10 ⁻⁴	1.1909	0.2718	9901	
Residual	180	6.8417*10 ⁻⁴				
Total	269					

d						
Source	df	MS	F	P	Unique Permutations	
Flow	2	3.9875*10 ⁵	53.477	0.0001	9947	
Density	2	1.4762*10 ⁶	197.97	0.0001	9951	
Flow x Density	4	71783	9.6269	0.0003	9953	
Time	5	16370	11.092	0.0001	9947	
Container (Flow, Density)	36	7456.6	5.0523	0.0001	9907	
Flow x Time	10	14368	9.7355	0.0001	9921	
Density x Time	10	9075	6.1489	0.0001	9934	
Flow x Density x Time	20	6139.5	4.1599	0.0001	9902	
Residual	180	1475.9				
Total	269					

e						
Source	df	MS	F	P	Unique Permutations	
Flow	2	2.8126*10 ⁰	598.46	0.0001	9951	
Density	2	2.4109*10 ⁵	51.299	0.0001	9950	
Flow x Density	4	57444	12.223	0.0001	9957	
Residual	50	4699.7				
Total	59					

f						
Source	df	MS	F	P	Unique Permutations	
Flow	2	1.8202	45.744	0.0001	9949	
Density	2	0.0049072	0.12333	0.8864	9960	
Flow x Density	4	0.0043251	0.1087	0.9783	9936	
Residual	45	0.03979				
Total	53					

c and d are Repeated Measures PERMANOVA analyses.
Significant differences (P<0.05) are in bold; df, degrees of freedom, MS, mean square

Effects of Environmental Parameters on Adult Growth and Gonad Production

DISTLM analyses indicated that 100% of all variation in adult SGR, LGR and GI could be explained by the first PCO axis; all variation could be attributed to differences between treatments. Temperature, pH, nitrate, and P_{CO_2} cumulatively accounted for 63.56%, 48.19% and 51.04% of the total variation for SGR, LGR and GI respectively. The individual parameter that accounted for the greatest total variation in SGR and LGR was P_{CO_2} (59.36% and 47.18% respectively). The individual parameter that accounted for the greatest total variation in GI was pH (48.16%).

Experiment 4: Partitioning Sources of Environmental Parameter Change in *Tripneustes gratilla* culture

Nitrate was detected in all treatments (Fig. 15a). There was no significant difference in nitrate concentration between the photoperiod phases (PERMANOVA, $F_{1,94} = 0.98761$, $P = 0.3217$), so replicates from different phases were analysed together. There were significant differences in nitrate concentration between both seawater exchange rates and treatments (Table 10a, followed by pair-wise post-hoc tests, Fig. 15a). At the High exchange rate, there was no significant difference in nitrate concentration between treatments and the control seawater. At the Low exchange rate, all treatments had significantly less nitrate than the control seawater, and were lowest in the treatments containing urchin faeces.

pH did not significantly differ between photoperiod phases (PERMANOVA, $F_{1,97} = 0.00089606$, $P = 0.9766$), so replicates from different phases were analysed together. pH was significantly affected by both seawater exchange rate and treatment (Table 10b). pH was lower in all treatments compared to the controls, regardless of exchange rate (Table 10b, followed by pair-wise post-hoc tests, Fig. 15b). pH was significantly lower in Low exchange rate treatments containing urchins compared to any High exchange rate treatments.

Total alkalinity did not significantly differ between the photoperiod phases (PERMANOVA, $F_{1,97} = 0.2523$, $P = 0.6284$), so replicates from different phases were analysed together. There were significant differences in total alkalinity between both exchange rates and treatments (Table 10c). Treatments containing *T. gratilla* had significantly less total alkalinity than those without urchins regardless of exchange rate (Table 10c, followed by pair-wise post-hoc tests, Fig. 15c). In treatments where urchins were present, total alkalinity concentrations were significantly lower in the Low exchange rate treatments than the High exchange rate treatments, and lower in the urchins only treatment compared to the combined treatment at the Low exchange rate. Treatments containing faeces only had significantly higher total alkalinity than controls.

P_{CO_2} did not significantly differ between photoperiod phases (PERMANOVA, $F_{1,93} = 0.36194$, $P = 0.5574$), so replicates from different phases were analysed together. P_{CO_2} was significantly affected by both seawater exchange rate and treatment (Table 10d). P_{CO_2} was higher in all treatments compared to the controls, regardless of exchange rate (Table 10d, followed by pair-wise post-hoc tests, Fig. 15d). P_{CO_2} was significantly higher in Low exchange rate treatments compared to High exchange rate treatments, except for the urchin only treatment where P_{CO_2} was equivalent between exchange rates.

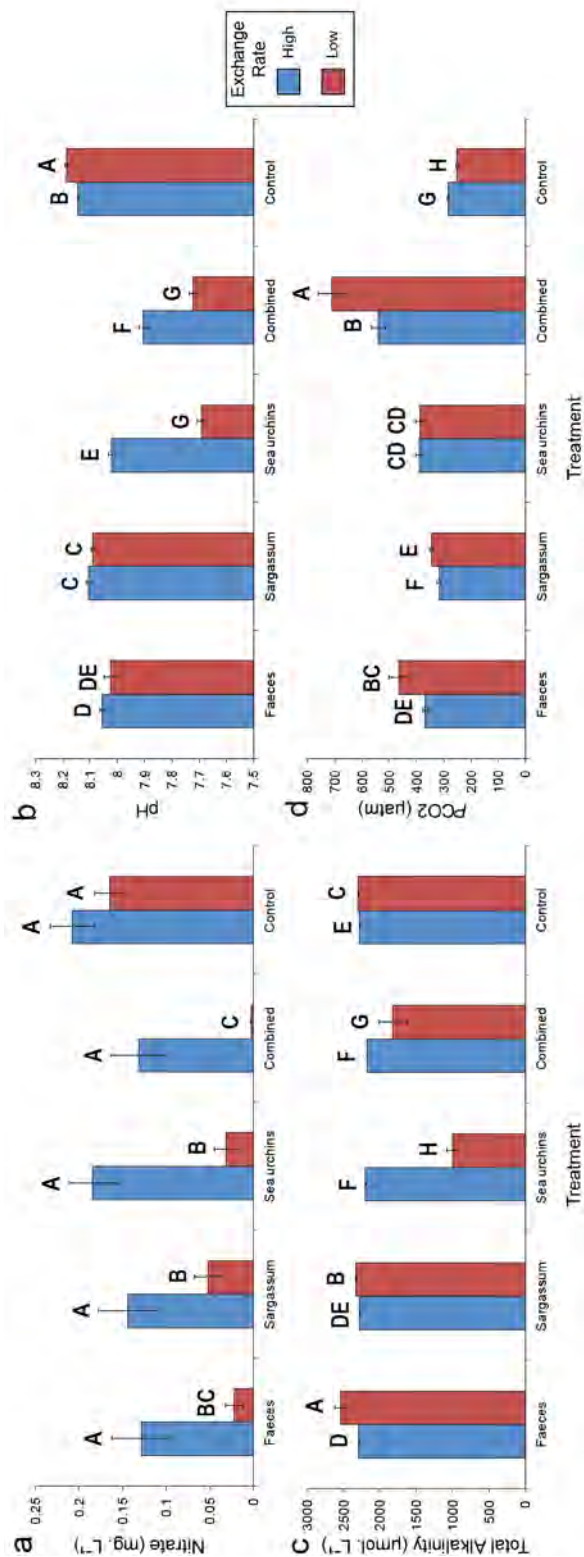


Figure 15: The effects of seawater exchange rate and presence of *Sargassum* spp., *Tripneustes gratilla* and *T. gratilla* faeces on (a) nitrate concentration, (b) pH, (c) total alkalinity and (d) carbon dioxide concentration (P_{CO_2}) in a flow-through seawater system. Combined = Faeces + Sargassum + Urchins, Control = Seawater only. Capitalised letters denote statistical differences between treatments (Table 6a,b,c,d respectively; followed by post-hoc pair-wise test, $P < 0.05$). Data are means $\pm$ S.E.

Table 10: PERMANOVA analyses examining the effects of seawater exchange rate and presence of *Sargassum* spp., *Tripneustes gratilla* and *T. gratilla* faeces on (a) nitrate concentration, (b) pH, (c) total alkalinity and (d) carbon dioxide concentration (P_{CO2}) in a flow-through seawater system

a					
Source	df	MS	F	P	Unique Permutations
Flow	1	0.25999	47.215	0.0001	9823
Treatment	4	0.040906	7.4287	0.0001	9944
Flow x Treatment	4	0.0079788	1.449	0.2277	9953
Residual	85	0.0055065			
Total	94				

b					
Source	df	MS	F	P	Unique Permutations
Flow	1	0.26182	158.38	0.0001	9851
Treatment	4	0.46449	280.99	0.0001	9941
Flow x Treatment	4	0.11183	67.65	0.0001	9946
Residual	88	1.6531×10^{-3}			
Total	97				

c					
Source	df	MS	F	P	Unique Permutations
Flow	1	1.6639×10^8	86.218	0.0001	9838
Treatment	4	2.3804×10^5	123.35	0.0001	9950
Flow x Treatment	4	1.8045×10^6	93.503	0.0001	9934
Residual	88	19299			
Total	97				

d					
Source	df	MS	F	P	Unique Permutations
Flow	1	62815	11.693	0.0007	9839
Treatment	4	3.631×10^5	67.593	0.0001	9952
Flow x Treatment	4	32075	5.971	0.0001	9958
Residual	84	5371.9			
Total	93				

Significant differences ($P < 0.05$) are in bold; df, degrees of freedom, MS, mean square

Discussion

A lack of data on optimal environmental conditions for juvenile and adult sea urchins is an important impediment to the establishment of *T. gratilla* aquaculture ventures in Australia. This study found that temperature, density and seawater exchange rate were important determinants of growth, survival and gonad production in juvenile and adult *T. gratilla*. Growth, survival and gonad production of *T. gratilla* was highest at 26–28°C, corresponding with optimal temperature ranges for larval and settlement phases (Rahman et al., 2009; Mos et al., 2011). *T. gratilla* can be grown at exceptionally high densities with relatively low seawater exchanges. Variation in growth and gonad production in response to varying density and seawater exchange rates were correlated with changes in environmental conditions in these treatments. In particular, pH was demonstrated to have the highest influence on growth rates and gonad production. Extrapolation of the results presented here will allow modelling of the commercial viability of *T. gratilla* culture in Australia. The growth rates of the sea urchins in this study are higher than any other published growth rates of sea urchins in a commercial scale setting. This highlights the great potential of this species for commercial aquaculture.

The effect of temperature on growth followed the typical non-linear pattern common to all ectotherms (Huey & Kingsolver, 1989). Growth of *T. gratilla* increased with increasing temperature until a thermal threshold was reached at approximately 29°C, after which there was a rapid reduction in growth. Optimal temperatures were similar to those found for the growth of larval and early juvenile *T. gratilla* (Rahman et al., 2009; Mos et al., 2011). Accordingly, it is recommended that the optimal culture temperature for *T. gratilla* is approximately 26–28°C. This temperature range was used in all subsequent experiments.

The effects of temperature on growth of juvenile and adult sea urchins in an aquaculture environment has surprisingly not been tested for many other species, and has been highlighted as an important knowledge gap (Azad et al., 2010). However, the majority of sea urchins species that have been investigated for their potential in aquaculture are from temperate climates where lower optimal growth temperatures result in slower growth than found here for *T. gratilla* (Lawrence & Bazhin, 1998). The high optimal growth temperature for *T. gratilla* indicates that this species would be suited to cultivation in tropical Northern areas of Australia, and would require some system of water heating in Southern areas. Currently, aquaculture in tropical Australia has focused on prawns, pearl oysters and a few species of fin fish. *T. gratilla* culture may provide an additional option for diversifying aquaculture operations.

Density of sea urchins and the rate of seawater exchange in the culture system both had a significant impact on the growth rate of juveniles and adults, and on the production of gonads in adults. Unsurprisingly, growth rates and gonad production rates were highest in high flow conditions and lower as the density increased. High exchange rates are important to flush waste materials from culture tanks, and increased densities increase the accumulation of these wastes. It is difficult to make a recommendation of optimal flow rates and densities for commercial production of *T. gratilla* based on these results. These decisions will need to be based upon models of the relative cost of increased exchange rates in a particular system traded off against the benefits and costs of variation in densities on growth and gonad production. It is clear from this study that the relative benefits of increased flow and decreased culture density will vary between the juvenile and adult phases of production. Importantly, the results of this study provide the data that could be used as an economic model for the production of sea urchins in a given aquaculture system.

pH of the culture water played by far the largest role in determining the growth and gonad production in *T. gratilla*. The effect of pH on growth of sea urchins is complicated because the concentration of carbon dioxide dissolved in sea water (expressed here as P_{CO_2}) and the availability of carbonate ions covaries with pH. As pH decreases, the availability of carbonate ions decreases, but the concentration of carbon dioxide increases. These three factors can have a negative impact on growth of sea urchins. High P_{CO_2} reduces growth because increased physiological concentrations of CO_2 (hypercapnia) have a narcotising effect. Reduced carbonate ion availability also has the potential to reduce growth in calcifying animals such as sea urchins because carbonate ions are an essential component of their

shells. As the availability of carbonate ions decreases, it is thought that sea urchins must expend more energy to form their tests reducing growth rates. Finally, low pH may also reduce growth because animals need to maintain internal pH within narrow ranges for optimal growth. Sea urchins are thought to be poor at regulating internal pH when environmental pH is variable.

The pH and associated parameters described above could have been altered in the sea urchin culture water in these experiments by two mechanisms. Respiration by the sea urchins could increase the concentration of dissolved carbon dioxide, reducing pH. The formation of calcium carbonate tests by the sea urchins strips carbonate ions from the seawater, also reducing the pH of the culture water. It is difficult to determine the extent to which respiration is causing change in these experiments, but indications of the role of calcification can be obtained from measuring total alkalinity. Total alkalinity is a measure of the concentration of carbonate species in water that is independent of pH or carbon dioxide concentration. Total alkalinity was reduced in all treatments compared to the ambient seawater supplied to the experiments. In the final experiment, total alkalinity was only reduced in the presence of sea urchins. This indicates that the sea urchins are stripping carbonate ions from water as they grow in the process of forming their calcium carbonate tests. From these results, decreased availability of carbonate ions and the resultant reduction in pH appears to be slowing growth.

Typically, low exchange rates and high densities have negative effects on animals in aquaculture systems due to reduced oxygen concentrations and the increase in nitrogenous wastes. In this study, oxygen concentration was unchanged at all flows and densities, and was always above 6.00mg.L^{-1} , or greater than 90% concentration. It appears maintaining oxygen concentration will not be an important factor in sea urchin cultivation. This is most likely because the low metabolic rate of sea urchins means they have low oxygen consumption compared to other aquaculture species (e.g. fin fish).

Ammonia and nitrite were generally not present at detectable levels ($>0.00\text{mg/L}$) in the culture water in this study. Surprisingly, reduced exchange rates reduced the concentration of nitrate compared to ambient concentrations detected in the seawater supplied to experiments. Reduced nitrate concentrations were possibly due to the uptake of nitrate by seaweed or by microbial action, as the final experiment could not partition between sea urchins, seaweed or faeces as the cause of the nitrate reduction. The lack of impact of nitrogenous waste in sea urchin culture is most probably because they feed upon seaweeds that have lower nitrogen contents than most feeds used for prawn and fish aquaculture. This has the added benefit that nitrogenous wastes from sea urchin aquaculture operations are unlikely to be a concern, whereas it is a serious regulatory issue for many aquaculture species.

Feeding rates and FCE rates were affected by culture density and seawater exchange rate. The sea urchins consumed more at high exchange rates and low densities, and also converted this food to body mass more efficiently at high exchange rates and low densities. There were large differences in the food consumption rates and FCEs between juvenile and adult urchins. Juvenile sea urchins ate approximately 8% of their body weight in seaweed per day and converted approximately 60% of this to body weight. However, adult sea urchins ate much less per day; approximately 5% of their body mass. They were also poorer at converting this food into body mass with a FCE of approximately 30%. These differences are not surprising as juveniles have higher inherent growth rates as they allocate a lower proportion of energy to maintenance. Adult urchins also direct some proportion of their energy intake into the production of energetically expensive gonads. Typically, FCEs for aquaculture species would be expected to be over 100%. It is important to note that sea urchins are herbivorous and are expected to have lower conversion efficiencies. However, this should be more than offset by the lower cost of producing low protein herbivore feeds. One reason for the low FCEs recorded in this study was because they were calculated based upon wet weight of seaweed, not dried weight as calculated for dried commercial finfish diets.

The growth rates reported here are higher than for any other species of sea urchin considered for aquaculture (Lawrence & Bazhin, 1998). This supports *T. gratilla* as an ideal candidate for aquaculture as growth rates recorded here were also maintained under high densities in both the juvenile and adult phases. Extrapolating from the data presented in this study, it appears that *T. gratilla* can be grown from weaning onto adult diet (TD 10mm) to harvest in approximately 8-10 months.

Recommendations

This report recommends *Tripneustes gratilla* as an ideal candidate for aquaculture. Its exceptionally high growth rates and ability to maintain high growth, survival and gonad production under extremely high densities makes it an attractive proposition for culture in Australia.

It is recommended that the optimum culture temperature for this species is between 26-28°C.

It is recommended that any prospective producer of *T. gratilla* should utilise the data in this report to construct models to investigate the economic viability of such an operation. Any such analysis would need to incorporate site specific information and relative pricing of seawater exchange rates, tank space and market prices for gonads. Because of this, this report does not make any specific recommendations regarding these parameters.

This report further recommends that several important hurdles to the commercial production of *T. gratilla* remain. These include:

- A bottle-neck in the survival of newly settled juveniles (see Mos et al., 2011).
- Establishment of protocols to allow uniform gonad conditioning of adults prior to harvest.
- Creation and production of cost-effective diets that optimise growth.

These problems will need to be addressed before the establishment of a sea urchin aquaculture industry in Australia can occur.

References

- Anderson, M.J., 2001. A new method for non-parametric multivariate analysis of variance. *Austral Ecology*. 26, 32-46.
2006. Distance-Based Tests for Homogeneity of Multivariate Dispersions. *Biometrics*. 62, 245-253.
- Anderson MJ, Gorley RN, Clarke KR (2008) PERMANOVA+ for Primer: Guide to software and statistical methods. PRIMER-E Ltd, Plymouth, 244pp.
- Andrew, N.L., Agatsuma, Y., Ballesteros, E., Bazhin, A.G., Creaser, E.P., Barnes, D.K.A., Botsford, L.W., Bradbury, A., Campbell, A., Dixon, J.D., Einarsson, S., Gerring, P.K., Hebert, K., Hunter, M., Hur, S.B., Johnson, C.R., Juinio-Menez, M.A., Kalvass, P., Miller, R.J., Moreno, C.A., Palleiro, J.S., Rivas, D., Robinson, S.M.L., Schroeter, S.C., Steneck, R.S., Vadas, R.L., Woodby, D.A., Xiaoli, Z., 2002. Status and management of world sea urchin fisheries. *Oceanography and Marine Biology*. 40, 343-425.
- Azad, A.K., McKinley, S., Pearce, C.M., 2010. Factors influencing the growth and survival of larval and juvenile echinoids. *Reviews in Aquaculture*. 2, 121-137.
- Azad, A.K., Pearce, C.M., McKinley, R.S., 2011. Effects of diet and temperature on ingestion, absorption, assimilation, gonad yield, and gonad quality of the purple sea urchin (*Strongylocentrotus purpuratus*). *Aquaculture*. 317, 187-196.
- Basuyaux, O., Blin, J.L., 1998. Use of maize as a food source for sea urchins in a recirculating rearing system. *Aquaculture International*. 6, 233-247.
- Bottger, S.A., Devin, M.G., Walker, C.W., 2006. Suspension of annual gametogenesis in North American green sea urchins (*Strongylocentrotus droebachiensis*) experiencing invariant photoperiod - Applications for land-based aquaculture. *Aquaculture*. 261, 1422-1431.
- Cook, E.J., Kelly, M.S., McKenzie, J.D., 1998. Somatic and gonadal growth of the sea urchin *Psammechinus miliaris* (Gmelin) fed artificial salmon feed compared with a macroalgal diet. *Journal of Shellfish Research*. 17, 1549-1555.
- Daggett, T.L., Pearce, C.M., Tingley, M., Robinson, S.M.C., Chopin, T., 2005. Effect of prepared and macroalgal diets and seed stock source on somatic growth of juvenile green sea urchins (*Strongylocentrotus droebachiensis*). *Aquaculture*. 244, 263-281.
- Daggett, T.L., Pearce, C.M., Robinson, S.M.C., 2006. A comparison of three land-based containment systems for use in culturing green sea urchins, *Strongylocentrotus droebachiensis* (Muller) (Echinodermata: Echinoidea). *Aquaculture Research*. 37, 339-350.
- Devin, M.G., 2002. Land-based echinoculture: a novel system to culture adult sea urchins. in: Yokota, Y., Matranga, V., Smolenicka, Z. (Eds.), *The Sea Urchin: from Basic Biology to Aquaculture*. A. A. Balkema Publishers, Lisse, pp. 145-159.
- Devin, M.G., Peacock, R.J., Stence, H.D., 2004. Development of grow-out techniques for juvenile sea urchins *Strongylocentrotus droebachiensis*. in: Lawrence, J.M., Guzman, O. (Eds.), *Sea Urchins: Fisheries and Ecology*. DEStech Publications, Lancaster, pp. 246-254.
- Dworjanyn, S.A., Pirozzi, I., Liu, W.S., 2007. The effect of the addition of algae feeding stimulants to artificial diets for the sea urchin *Tripneustes gratilla*. *Aquaculture*. 273, 624-633.

- Grosjean, P., Spirlet, C., Gosselin, P., Vaitilingon, D., Jangoux, M., 1998. Land-based, closed-cycle echiniculture of *Paracentrotus lividus* (Lamarck) (Echinoidea : Echinodermata): A long-term experiment at a pilot scale. *Journal of Shellfish Research*. 17, 1523-1531.
- Hammer, H., Hammer, B., Watts, S., Lawrence, A., Lawrence, J., 2006. The effect of dietary protein and carbohydrate concentration on the biochemical composition and gametogenic condition of the sea urchin *Lytechinus variegatus*. *Journal of Experimental Marine Biology and Ecology*. 334, 109-121.
- Huey, R.B., Kingsolver, J.G., 1989. Evolution of thermal sensitivity of ectotherm performance. *Trends Ecol. Evol.* 4, 131-135.
- Juinio-Menez, M.A., Bangi, H.G.P., Malay, M.C.D., 2008. Effect of Type of Feed, Stocking Density and Grow-out site on Gonad Index, Growth and Survivorship of Cultured Sea Urchin (*Tripneustes gratilla*). *Philippine Agricultural Scientist*. 91, 439-449.
- Kelly, M.S., 2002. Survivorship and growth rates of hatchery-reared sea urchins. *Aquaculture International*. 10, 309-316.
- Lawrence, J.M., Agatsuma, Y., 2001. Ecology of *Tripneustes*. in: Lawrence, J.M. (Ed.), *Edible Sea Urchins: Biology and Ecology*. Elsevier Science, Amsterdam, pp. 499-520.
- Lawrence, J.M., Bazhin, A., 1998. Life-history strategies and the potential of sea urchins for aquaculture. *Journal of Shellfish Research*. 17, 1515-1522.
- McBride, S.C., Lawrence, J.M., Lawrence, A.L., Mulligan, T.J., 1998. The effect of protein concentration in prepared feeds on growth, feeding rate, total organic absorption, and gross assimilation efficiency of the sea urchin *Strongylocentrotus franciscanus*. *Journal of Shellfish Research*. 17, 1563-1570.
- Miller, B.A., Emlet, R.B., 1999. Development of newly metamorphosed juvenile sea urchins (*Strongylocentrotus franciscanus* and *S. purpuratus*): Morphology, the effects of temperature and larval food ration, and a method for determining age. *Journal of Experimental Marine Biology and Ecology*. 235, 67-90.
- Mos, B., Cowden, K.L., Nielsen, S.J., Dworjanyn, S.A., 2011. Do Cues Matter? Highly Inductive Settlement Cues Don't Ensure High Post-Settlement Survival in Sea Urchin Aquaculture. *PLoS ONE*. 6, e28054.
- Pearce, C.M., Daggett, T.L., Robinson, S.M.C., 2002a. Effect of binder type and concentration on prepared feed stability and gonad yield and quality of the green sea urchin, *Strongylocentrotus droebachiensis*. *Aquaculture*. 205, 301-323.
- Pearce, C.M., Daggett, T.L., Robinson, S.M.C., 2002b. Effect of protein source ratio and protein concentration in prepared diets on gonad yield and quality of the green sea urchin, *Strongylocentrotus droebachiensis*. *Aquaculture*. 214, 307-332.
- Pearce, C.M., Williams, S.W., Yuan, F., Castell, J.D., Robinson, S.M.C., 2005. Effect of temperature on somatic growth and survivorship of early post-settled green sea urchins, *Strongylocentrotus droebachiensis* (Muller). *Aquaculture Research*. 36, 600-609.
- Pierrot, D., Lewis, E., Wallace, D.W.R., 2006. MS Excel Program Developed for CO2 System Calculations. in: Center, O.C.-a.C.D.I.A. Oak Ridge National Laboratory, U.S. Department of Energy, Oak Ridge, Tennessee, pp. doi:10.3334/CDIAC/otg.CO3332SYS_XLS_CDIAC3105a.

- Rahman, S., Tsuchiya, M., Uehara, T., 2009. Effects of temperature on hatching rate, embryonic development and early larval survival of the edible sea urchin, *Tripneustes gratilla*. *Biologia*. 64, 768-775.
- Sheppard Brennand, H., Soars, N., Dworjanyn, S.A., Davis, A.R., Byrne, M., 2010. Impact of Ocean Warming and Ocean Acidification on Larval Development and Calcification in the Sea Urchin *Tripneustes gratilla*. *PLoS ONE*. 5, e11372.
- Shimabukuro, S., 1991. *Tripneustes gratilla*. in: Shokita, S., Yamaguchi, M., Masashi, M. (Eds.), *Aquaculture in Tropical Areas*. Midori Shobo, Tokyo, pp. 313-328.
- Siikavuopio, S.I., Dale, T., Christiansen, J.S., Nevermo, I., 2004a. Effects of chronic nitrite exposure on gonad growth in green sea urchin *Strongylocentrotus droebachiensis*. *Aquaculture*. 242, 357-363.
- Siikavuopio, S.I., Dale, T., Foss, A., Mortensen, A., 2004b. Effects of chronic ammonia exposure on gonad growth and survival in green sea urchin *Strongylocentrotus droebachiensis*. *Aquaculture*. 242, 313-320.
- Siikavuopio, S.I., Dale, T., Mortensen, A., 2007a. The effects of stocking density on gonad growth, survival and feed intake of adult green sea urchin (*Strongylocentrotus droebachiensis*). *Aquaculture*. 262, 78-85.
- Siikavuopio, S.I., Dale, T., Mortensen, A., Foss, A., 2007b. Effects of hypoxia on feed intake and gonad growth in the green sea urchin, *Strongylocentrotus droebachiensis*. *Aquaculture*. 266, 112-116.
- Siikavuopio, S.I., Mortensen, A., Dale, T., Foss, A., 2007c. Effects of carbon dioxide exposure on feed intake and gonad growth in green sea urchin, *Strongylocentrotus droebachiensis*. *Aquaculture*. 266, 97-101.
- Siikavuopio, S.I., Mortensen, A., Christiansen, J.S., 2008. Effects of body weight and temperature on feed intake, gonad growth and oxygen consumption in green sea urchin, *Strongylocentrotus droebachiensis*. *Aquaculture*. 281, 77-82.

Potential for the Commercial Culture of the Tropical Sea Urchin *Tripneustes gratilla* in Australia

by B. Mos, K. L. Cowden and S.A. Dworjanyn

Pub. No. 12/O52

The decline in sea urchin fisheries production has seen increasing interest in the commercial culture of sea urchins, including the fast-growing tropical *Tripneustes gratilla*. However, the establishment of commercial sea urchin culture ventures is hampered by a lack of data on optimal culture conditions for sea urchins. The information presented in this report provides data to bridge this knowledge gap and represents an important step forward in the realisation of commercial sea urchin aquaculture.

This report is targeted at commercial entities interested in diversifying into sea urchin aquaculture and also contributes to the growing global literature on sea urchin aquaculture, and may therefore be of interest to aquaculture researchers.

RIRDC is a partnership between government and industry to invest in R&D for more productive and sustainable rural industries. We invest in new and emerging rural industries, a suite of established rural industries and national rural issues.

Most of the information we produce can be downloaded for free or purchased from our website <www.rirdc.gov.au>.

RIRDC books can also be purchased by phoning 1300 634 313 for a local call fee.



RURAL INDUSTRIES
Research & Development Corporation

Phone: 02 6271 4100

Fax: 02 6271 4199

Bookshop: 1300 634 313

Email: rirdc@rirdc.gov.au

Postal Address: PO Box 4776,
Kingston ACT 2604

Street Address: Level 2, 15 National Circuit,
Barton ACT 2600

www.rirdc.gov.au